The Book of Z-Spin Cosmology — Light Edition(OS for AI)

Cross-Reference Hub for the 107-Paper Z-Spin Corpus

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Edition: Light v3.5 (compiled May 2026; v3.0–v3.4 May 2026)
Source corpus: 108 source papers across 8 themes (F / M / S / U / A / Q / QC / T)
Supersedes: The Book of Z-Spin Cosmology v3.4 — Light Edition (May 2026, 107-paper coverage; absorbed ZS-T5, M34, M35, T6, T7, T8). v3.5 absorbs 1 post-v3.4 paper (ZS-M36) → 108-paper coverage. (Earlier supersession chain: v3.3 added PART XI; v3.4 added 6 papers from v3.3.)
GitHub: https://github.com/KennyKang-git/zspin

Verification Summary

Verification: cumulative ~1,800+ PASS across the 107-paper corpus | Zero Free Parameters | A = 35/437, Q = 11, (Z, X, Y) = (2, 3, 6) sole geometric inputs | Anti-numerology controls: 5670 × 29 = 164,430 candidate scan + per-paper Monte Carlo (≥ 500,000 samples typical) | Pre-registered falsification gates: ~250 across all themes; Seven critical near-term ★ gates timeline preserved.

§0. Abstract

The Book v3.0 Light Edition is the canonical cross-paper navigation hub for the 101-paper Z-Spin Cosmology corpus as of May 2026. It supersedes v2.0 Light by absorbing the thirty post-v2.0 papers (papers 72–101) — most of which extend the Foundations theme (ZS-F9 through ZS-F18), the Math Spine Riemann-Hypothesis programme (ZS-M18 through ZS-M33), and add the Twin-Reuleaux EM-duality ZS-S15 in the Standard Model theme, the contracting-sector ZS-A8 and Banach–Tarski origin ZS-A9 in the Astrophysics theme, and the Cosmos-Human isomorphism ZS-T4 in the Translational theme. All chapter bodies remain compressed to single-line pointers of the form → ZS-XX §Y.Z [N/N PASS]; full derivations live in the source papers. The Light Edition's unique contributions are: (i) the post-v2.0 dependency graph integrating the 30 new papers, (ii) the expanded Foundations closure ring F0 → F1 → F2 → F5 → F8 → F9 → F10 → F11 → F12 → F13 → F14 → F15 → F16 → F17 → F18, (iii) the Math Spine RH-bridge programme map (M22–M33), (iv) the extended Glossary keyword index (~110 entries), and (v) the Seven Critical Near-Term ★ Gates timeline. The Light Edition does NOT replace any source paper for archival purposes; it is the LLM-corpus navigation aid and the v3.0 composition skeleton.

Epistemic Status Legend

Status	Definition

PROVEN	Mathematical theorem; standard math alone, machine-verifiable.
DERIVED	Z-Spin action + standard physics, zero free parameters.
DERIVED-CONDITIONAL	DERIVED conditional on listed axiom set or upstream theorem.
DERIVED-CONDITIONAL strong	DERIVED-CONDITIONAL with structural over-determination on the conditioning step.
DERIVED-under-P6 / DERIVED-under-Regge	Conditional on specific framework (ZS-F5 P6 or Regge calculus).
DERIVED-interpretation	Synthetic reading combining PROVEN corpus theorems without new axioms.
DERIVED-CANDIDATE	Mathematical content PROVEN; Z-Spin internal placement HYPOTHESIS-strong.
VERIFIED	Numerical / computational / empirical confirmation at stated precision.
TESTABLE	Pre-registered prediction with explicit falsification protocol.
TESTABLE-LONG	Pre-registered prediction with derivation-or-retraction time horizon ≥ 5 years.
HYPOTHESIS-strong	Multiple independent structural anchors; promotion path documented.
HYPOTHESIS / HYPOTHESIS-weak	Motivated conjecture; partial derivation chain.
OBSERVATION	Empirical regularity; anti-numerology controlled; structural origin pending.
LOCKED	Core constant fixed in upstream paper; no downstream paper may modify.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework's scope.
OPEN	Recognized gap honestly registered for future work.
RETRACTED	Earlier claim withdrawn after falsification.
RETRACTED-in-session	Hypothesis proposed during free-exploration and explicitly retracted before exit.
BOOTSTRAP-HYPOTHESIS	Meta-logical foundation (B0); not derivable within the framework.
BOOTSTRAP-PROGRAMME	Status reserved for the founding axiom set after ZS-F0 v1.0(R) Stage 1–5 closures.
STRUCTURAL INSIGHT	Exact mathematical identity; physical necessity not yet established.
FMD-SPEC / TRANSLATED / CANDIDATE	Engineering tags used in the QH/QS/QC translational papers.

Purpose of the Light Edition

The full v3.0 (when compiled) consolidates the 101-paper corpus into chapter bodies that reproduce derivations already present verbatim in the source papers. For LLM corpus-wide

exploration, this redundancy is wasteful. The Light Edition removes the duplication and retains only what The Book contributes uniquely:
(1) The cross-paper dependency graph (Source Paper Interface Tables, PARTS 0–VII).
(2) Standalone HYPOTHESIS / theorem-anchor sections cited by individual papers (§8.4i Spinor–Descartes–Euler, §8.4g Y-sector R_micro, §8.4k Attractor-Flatness, §VI.2 Limitation Catalog Row 10, §18.10 Horizon Spinor mirror).
(3) Cumulative Master Falsification Registry — Seven Critical Near-Term ★ Gates timeline.
(4) Standardized v2.0 / v3.0 terminology contract (Z-sector / Z-Spin / Z-bias field; the v2.0 contract is preserved verbatim and v3.0 adds the Stage / Operator / Field separation explicitly via ZS-F18 Bridge Four).
(5) Glossary keyword index (~110 entries, one line each).
(6) Integration map for the 30 post-v2.0 papers (papers 72–101) consolidated into the appropriate PART/Chapter slots.

Chapter bodies that reproduce source-paper derivations are replaced by single-line pointers of the form → ZS-XX §Y.Z [N/N PASS]. When derivation detail is needed, retrieve the source paper directly. When corpus-wide context is needed, consult this Light Edition.

PART 0 — Overture: Why Geometry?

§0. Abstract

PART 0 establishes the framework's overture in three movements. Chapter 0.1 presents the Five Crises of modern physics motivating a geometric origin of fundamental constants. Chapter 0.2 lays out the methodological contract — standardized terminology, anti-numerology discipline, epistemic classification, pre-registered falsification gates. Chapter 0.3 traces the ontological bootstrap chain B0 → B3 → A. Concludes with FFPP of ZS-F0 v1.0(Revised) compressing all numerical content to $W_0(-i\pi/2)$, conditional on five-axiom set $\mathscr{A}$ = {Existence, Self-reference, Algebra, Non-triviality, Unitarity}. NEW v3.0: the Five-Axiom Meta-Structure of ZS-F18 v2.1 reads the corpus's encounter with the seven Clay Millennium Problems and the five Crises as outputs of the same axiom set, and registers Bridge Four (Commutator Generator) as the structural reduction of dim(X) = 3 to dim(Z) = 2.

§0.1 The Five Crises of Modern Physics

Crisis 1 — Hubble Tension: H_0 ~5σ disagreement (SH0ES vs Planck). Resolved via 2A modification → ZS-A2, ZS-S3, ZS-F12R (factor 2 origin).
Crisis 2 — Growth Tension (S_8): Z-Spin predicts $S_8 \approx 0.777$ (face-counting v2.0) → ZS-A2, ZS-A5.
Crisis 3 — Hierarchy Problem: Geometric origin via A → ZS-S4 §6.7, §6.12, §6.16; Light §8.4k.
Crisis 4 — Baryon Asymmetry: $\eta\_B = (6/11)^{35}$ → ZS-S5, ZS-S6, ZS-U7.
Crisis 5 — Measurement Problem: Born rule DERIVED from L_XY ≡ 0 → ZS-Q1, ZS-Q2.

§0.2.2 Standardized v3.0 Terminology [CONTRACT — preserved verbatim from v2.0; v3.0 elaboration in italics]

Term	Role	Meaning
Z-sector	Stage / Kinematics	2D Planck-scale boundary surface mediating between X (3D macroscopic) and Y (6D microscopic) sectors. dim(Z) = 2 PROVEN from gauge constraint on Q = 11 register (ZS-F5). Geometric arena where transduction occurs.
Z-Spin	Operator / Dynamics	Dynamical operator on Z-sector. Mediates between X (Up & Down, ±π, time-points) and Y (In & Out, slog(i) & ∞i, space-points) via i-tetration self-referential rotation T(z) = i^z (HSI Theorem, ZS-M1 §1.1). The action driving transduction.
Z-bias field Φ	Field / DOF	Complex scalar Φ(x) = ρ(x)exp(iθ(x)) ∈ ℂ on Z-sector. Radial mode ρ = |Φ|, angular mode θ ∈ [0, 2π). U(1)_Z derived via Frobenius (ZS-F1 §2.3). Degree of freedom on which Z-Spin acts.
Z-Anchor	Topological core	Bogomolnyi vortex core, |Φ| = 0 locus on Z-sector. → ZS-A6, ZS-F1 §5.2 PROVEN.
Z-bottleneck	Channel constraint	L_XY ≡ 0 PROVEN forces all X↔Y traffic through Z; channel capacity ≤ ln 2. → ZS-Q7.
Z-channel	Structural conduit	Operational pathway for X↔Y mediation across the 2D boundary. → ZS-Q1.
Z-Spin mediation	Action of the operator	Mediation is Z-Spin's action; the agent is the operator (Spin), not the stage (sector). Replaces 'Z-mediation' / 'Z-sector mediation' / 'Z-mediated coupling' / 'Z-mediated CPTP channel' throughout the corpus.
Z-EFT	Effective theory	Distinct concept (effective field theory built on Z-bias dynamics); preserved name.

[STRUCTURAL DEFINITION] Z-sector (stage, kinematics) / Z-Spin (operator, dynamics) / Z-bias field Φ (field, DOF). Source: ZS-F1 §2-3, ZS-F5, ZS-M1 §1.1, Glossary v1.0.0 §I. v3.0 elaboration: ZS-F18 v2.1 §6 Bridge Four (Commutator Generator) makes the kinematic / dynamical / field-DOF separation an explicit structural theorem — twin-Reuleaux pair (R_1, R_2) realizes (V_XZ, V_ZY); their Lie commutator [J_{R_1}, J_{R_2}] = J_S generates the third axis of dim(X) = 3.

§0.3 The Ontological Bootstrap Chain B0 → A

B0 (Existence) → B1 (Self-reference) → B2 (Algebra: ℂ via Lawvere/Frobenius) → B3 (Hyperoperation: tetration) → A = 35/437. F-BOOT 9/9 closure: 6 DERIVED, 2 structural-PASS, 1 DERIVED+TESTABLE. → ZS-F0 v1.0(Revised) §11–§13 [51/51 PASS].
FFPP (Theorem 13.3): entire Z-Spin numerical content compresses to $W_0(-i\pi/2)$, conditional on $\mathscr{A}$ = {Existence, Self-reference, Algebra, Non-triviality, Unitarity}. → ZS-F0 §13, Theorem 13.3 + Corollary 13.4.
[OPEN] B0 itself is BOOTSTRAP-HYPOTHESIS — not derivable within framework. → ZS-F0 §0.3.7.

NEW v3.0: ZS-F18 v2.1 reads the same five-axiom set $\mathscr{A}$ as the structural reason Z-Spin naturally encounters (i) the seven Clay Millennium Problems and (ii) the five Crises of modern physics; the encounters are classified by depth (direct treatment / extensive programme / structural correspondence / natural extension domain). NON-CLAIM: encounters are not formal solutions. → ZS-F18 v2.1 §0, §6, §7.4 (Discrete-Continuous Resolution polarity), Appendix A.

§0.4 PART 0 Source Paper Map

Source paper	Role	Light pointer
ZS-F0 v1.0(Revised)	Bootstrap, FFPP, BV-BFV functor, $\mathbb{C}^{11}$ 3-layer	→ ZS-F0 [51/51 PASS]
ZS-F1 v1.0	Z-Spin Action, U(1)_Z, Z-bias field	→ ZS-F1 §2-3 [30/30 PASS]
ZS-F2 v1.0	A = 35/437 LOCKED, δ_X = 5/19, δ_Y = 7/23	→ ZS-F2 [PROVEN]
ZS-F5 v1.0	dim(Z) = 2 from Q = 11	→ ZS-F5 [PROVEN]
ZS-F8 v1.0(Revised)	NOT/AND duality, 2-channel handshake	→ ZS-F8 [27/27 PASS]
ZS-F18 v2.1 (NEW)	Five-Axiom Meta-Structure, Bridge Four, Discrete-Continuous Resolution	→ ZS-F18 [36/36 entry-checks]
ZS-M1 v1.0	i-tetration HSI Theorem, fixed point z*	→ ZS-M1 §1.1 [33/33 PASS]
ZS-M22 v1.1 (post-v2.0)	Five-Pillar Arithmetic-Dedekind Scaffold; Forcing chain underlay	→ ZS-M22 [60/60 PASS]
ZS-T2 v1.0	Anti-numerology MC scan, 5670×29	→ ZS-T2 [30/30 PASS]

PART I — Foundations: One Number, One Action

§0. Abstract

A = 35/437 from polyhedral curvature asymmetry; Q = 11 register and (Z, X, Y) = (2, 3, 6) doubly forced by gauge algebra (ZS-F5) and totient arithmetic (ZS-M19); Z-Spin action with Z-Anchor and twin-Reuleaux kinematics; i-tetration fixed point z* = 0.4383 + 0.3606i; FFPP compressing all content to $W_0(-i\pi/2)$; Operational Observer Coordinate closing Foundations as a six-paper ring; v3.0 extension to a thirteen-paper Foundations theme F0 → F1 → F2 → F5 → F8 → F9 → F10 → F11 → F12 → F13 → F14 → F15 → F16 → F17 → F18 with Möbius chronology, Stage-Torque Duality, Two-Protocol Theorem, Self-Referential Compression Equilibrium, and the Five-Axiom Meta-Structure as load-bearing additions.

Chapter 1 — A = 35/437 from Polyhedral Curvature

A = δ_X · δ_Y = (5/19)(7/23) = 35/437. δ-Uniqueness Theorem PROVEN. Three independent derivations: (i) Regge curvature asymmetry, (ii) ZS-F9R Truncation-Residue Identity (PROVEN: ρ_X = δ_X = 5/19, ρ_Y = δ_Y = 7/23, ρ_Z = 0; A = ρ_X · ρ_Y), (iii) ZS-F12R Layer Separation Hypothesis (factor 2 in 2e^A from tetrahedral V↔F dual orientation multiplicity μ_Tet = 2). → ZS-F2 §3, §11; ZS-F9R [44/44 PASS]; ZS-F12R [20/20 PASS].

Chapter 2 — Q = 11 and (Z, X, Y) = (2, 3, 6)

Doubly forced: (a) gauge dimension G = 12 = MUB(Q=11), (b) Forcing Theorem $(p-1)(q-1) = p$ has unique solution $(p,q) = (2,3)$ — the prime pair (Z, X) = (2, 3) is the only one for which $\varphi(Y^2) = G$ (T1, ZS-M19 §3.1). Y = ZX. $Q^2 - 1 = 120 = |I_h|$. → ZS-F5 [PROVEN]; ZS-M19 §3.1 [68/68 PASS]; arithmetic-Dedekind underlay → ZS-M22 v1.1 [60/60 PASS].

Chapter 3 — The Z-Spin Action

$S = \int d^4x \sqrt{(-g)}[(M\_P^2/2)(1+A|\Phi|^2)R - (M\_P^2/2)|\nabla \Phi|^2 - V(\Phi)] + S\_matter$. $V = (\lambda/4)M\_P^4(|\Phi|^2-1)^2$. $L\_XY \equiv 0$ PROVEN. $\lambda\_vac = 2A^2$ IR-stable. → ZS-F1 [30/30 PASS]; ZS-F11 NC catalog.

Chapter 4 — i-Tetration Fixed Point

$z^* = 0.43828 + 0.36059i$, $x^* = 0.438283$. Five Locking Conditions. $\pi/2$ phase budget. Polygon-Tetration family stable for $n \geq 4$ ($n\_c = 3.204$). NEW v2.0/v3.0: ZS-F10 (3 time coords unified at DERIVED-CONDITIONAL strong; $\Delta\nu/\Delta n = 2A/\pi$ conversion identity), ZS-F15R (Time-Rotation Co-Definition + Stage-Torque Duality), ZS-M12 (Auto-Surgery, $\varepsilon\_min \approx 30.7$), ZS-F14 (Kepler-Lyapunov conjugate decomposition of twin-Reuleaux midpoint trajectory; F-F7.11 closed by Strategy B at DERIVED-CONDITIONAL [42/42 PASS]). → ZS-M1 §1.1 [PROVEN]; ZS-F10 [30/30 PASS]; ZS-F15R [35/35 PASS]; ZS-M12; ZS-F14 [42/42 PASS].

Addition 1 — Self-Reference and FFPP

Lawvere bridge (Th 11.1–11.4), Frobenius (Th 11.5), hyperoperation minimality (§11.6), NOT/AND duality, BV-BFV cobordism functor B_Z with Wilson loop $|Z(W)|^2 \approx 0.7948$, three-layer $\mathbb{C}^{11}$ decomposition (Th 9.1, 9.4), FFPP (Th 13.3) and ultimate compression to $W_0(-i\pi/2)$ (Cor 13.4). All conditional on B0. NEW v3.0: ZS-F17 v1.0 four-layer 50/50 equilibrium (truncated tetrahedron 4+4 + Reuleaux $\pi+\pi$ + Master Equation Term A=Term B + XOR E vs R) sharpens FFPP to the operational Self-Referential Information-Compression Equilibrium Theorem. → ZS-F0R [51/51 PASS]; ZS-F8R [27/27 PASS]; ZS-F17 [24/24 PASS].

Addition 2 — Operational Observer Coordinate (OOC) and Möbius Chronology

OOC closes ZS-M11 §H16 at register-theoretic level. Theorem F11.1: Observer Coordinate Decomposition (P1 inside Z, P2 higher-level event count, P3 orthogonal kinematic $|5\rangle$). Six explicit NON-CLAIMs against phenomenological consciousness reading. Sectoral Frame Equivalence (X-Frame ↔ Y-Frame). Foundations theme closes as a six-paper ring (F0-F1-F2-F5-F11-F13). NEW v3.0: ZS-F13 v1.0 Möbius Chronology Theorem completes the cyclic timeline with Y-frame metric trio (proper time, Y-Hubble radius $\ell\_P \times Y^2(1-2A) \approx 30.23$

ℓ_P, Y-contraction rate (1−2A)) and proves cycle-index unobservability (Theorem F13.2A DERIVED, F13.2B DERIVED-CONDITIONAL). The six-paper ring is now extended to the eight-paper Möbius ring. → ZS-F11 [38/38 PASS]; ZS-F13 §15.5g.5 [24/24 PASS].

Addition 3 — NEW v3.0 — Stage-Torque Duality and First-Rotation Closure

ZS-F15 v1.0(Revised) [35/35 PASS] closes NC-M12.4 (Q = A assumes $\dot{\theta}$ = A at Z-Telomere onset). The Mexican-Hat Bootstrap Event simultaneously instantiates radial mode, angular mode, and time evolution; $\dot{\theta}(0) \cdot \tau_P = \delta\varphi_{cell} = A$ is a unit-fixing identity, not a free parameter (Theorem F-rotor.3 DERIVED — central new contribution). The Stage-Torque Duality (Theorem F-rotor.5 DERIVED): Mexican-Hat stage from $V(\Phi)$ opens U(1)-symmetric S^1 vacuum manifold; A-holonomy torque from Wilson loop $\oint \omega = A$ on polyhedral defect manifold (Gauss-Bonnet on $O_h \times I_h$) provides rate-fixing torque. The same $\oint \omega = A$ produces both the Planck-scale $\delta\varphi_{cell} = A$ and the macroscopic Hubble ratio $H_0^{local}/H_0^{CMB} = \exp(A)$ (ZS-F3 §6 DERIVED), unifying physics across orders of magnitude. FFPP Sixth Axiom Extension (HYPOTHESIS-strong) registered.

Addition 4 — NEW v3.0 — Two-Protocol Theorem for Z-Sector Wilson Loop Measurement

ZS-F16 v1.0 [24/24 PASS, 32-test target in roadmap] resolves the WL-1/WL-2 implicit ambiguity by establishing two operationally distinct Z-sector Wilson loop protocols: Protocol (a) state-preparation on BFV anchor $|0\rangle_Z$ gives $P_a(n) = 0.7948^n \cdot \cos^2(n\cdot 129.45°)$ (quasi-revival pattern); Protocol (b) spectral characterization gives $P_b(n) = 0.7948^n$ (smooth exponential decay). Reconciliation by ergodic averaging since $\arg(\lambda) = 129.45°$ is irrational. Audit by sum rule 0.7948 + 0.2050 + 0.0001 = 0.9999. Z-block effective dissipation $\Gamma_Z \cdot T_{cycle} \approx 0.1149$. NEW sub-gates F-CTA-1a/b/c/d/multi-a/multi-b/-discriminator pre-registered for 2027+ quantum hardware testing.

Addition 5 — NEW v3.0 — The Five-Axiom Meta-Structure (ZS-F18)

ZS-F18 v2.1 (interpretive synthesis paper, 36/36 internal consistency) reads the Z-Spin corpus as the unique geometric framework that encounters (a) the seven Clay Millennium Problems and (b) the five Crises of modern physics simultaneously, with these encounters traced back to the same five-axiom set $\mathscr{A}$ = {Existence, Self-reference, Algebra, Non-triviality, Unitarity}. Bridge Four (Commutator Generator, NEW v2.0): two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair compose at 90° NOT-AND phase to generate the third axis $J_S = [J_{R_1}, J_{R_2}]$ by Lie commutator. The full sector decomposition (Z, X, Y) = (2, 3, 6) and register Q = 11 are therefore reread as outputs of the foundational dim(Z) = 2 alone. v2.1 sixth polarity row: Discrete-Continuous Resolution via X–Y Tiling Asymmetry (ZS-M6 §5.5 PROVEN; ZS-M17 Tiling Continuum Convergence DERIVED). Status: INTERPRETATION / DERIVED-interpretation strong. Twelve encounters classified by depth in Appendix A.

PART I Source Paper Map (v3.0 extended)

Theme	Papers	Pointer
Foundations [F]	F0R, F1, F2, F5, F8R, F9R, F10, F11, F12R, F13, F14, F15R, F16, F17, F18 (v2.1)	→ 01_Foundations (15 papers)
Math Spine [M]	M1, M3, M11, M12, M14, M19, M22 (v1.1), M36 (NEW v3.5)	→ 02_Math_Spine
Cross-link	S1, S15 (Twin-Reuleaux EM duality), Q-theme, A6/A7/A8/A9	various

PART II — Standard Model Completion

§0. Abstract

Three SM gauge couplings DERIVED at zero free parameters: $\alpha_s(M_Z) = 11/93 = 0.11828$ ($+0.31\sigma$), $\sin^2\theta_W = (48/91)\cdot x^* = 0.23118$ (-1.26σ), $\alpha_2 = 3/95$. Fine-structure $1/\alpha_{EM} = 137.0359$ (1.07 ppm) via McKay $c_4 = 4/13$. Yang-Mills mass gap structurally addressed: $\Lambda_{QCD} = 264$ MeV, $m(0^{++}) = 1.791$ GeV. Neutrino seesaw $m_D = m_e\cdot A = 40.93$ keV, $M_R = 33.50$ GeV, $|\theta|^2 = 1.49\times10^{-12}$. Inverted Ordering canonical. Cabibbo from icosahedral $\sigma_1/\sigma_2 = 17$. NEW v3.0: ZS-S15 [35/35 PASS] — twin-Reuleaux pair as the geometric realization of EM field duality, Maxwell vacuum equations as time-point/space-point handshake duality, SO(3)/SU(2) factor 2 from half-angle holonomy; ZS-M20 [50/50 PASS] — spectral tetrad sub-isotype assignment with closed-form lepton mass-ratio chain $\sigma_1/\sigma_2 = Q+Y = 17$, $\sigma_1/\sigma_3 = (Q+Y)^2\cdot G + (Q-Z) = 3477$ EXACT; ZS-M21 [33/33 PASS] — sextic invariant theory of the icosahedral Yukawa tensor.

§II-UNIQUE — Book-Standalone Hypotheses Cited by Source Papers

This section preserves Book content referenced by individual source papers as their primary citation. Removing these would orphan citation chains in ZS-A7, ZS-S4, ZS-S15, others.

§8.4i — Spinor–Descartes–Euler Identity [PROVEN]

Three classical theorems converge: Descartes $\Sigma\delta v = 4\pi$; Euler $\Sigma\delta v = 2\pi\cdot\chi$, $\chi(S^2) = 2$; ZS-M3 Theorem 5.1 $\dim(Z) = 2 = \chi(S^2)$ unique $j = 1/2$ invariant subspace. Unification: $\Sigma\delta v = 2\pi\cdot\chi = 2\pi\cdot\dim(Z) = 4\pi$ = spinor full-return period. Used in ZS-S7 §3 (Λ_{QCD}, $m(0^{++})$, 18/18 PASS); ZS-A7 §2.2 (Pillar 2 of Horizon Spinor); ZS-S15 §5 (Pillar IV: SO(3)/SU(2) factor 2).

§8.4g — Y-sector R_micro UV-IR Matching [HYPOTHESIS structural]

B-2 reinterpretation framework. Cross-cited by ZS-A7 §4.5 Corollary V (Higgs as micro radial-mode attractor) and ZS-S4 §6.12.8–§6.12.13 (B-2 repackaging). Cross-paper anchor for Y-sector micro UV-IR boundary condition.

§8.4k — Attractor-Flatness Theorem [HYPOTHESIS structural]

Compatible with ZS-A7 Corollaries I–IV. Two attractors: macro $V(|\Phi|=1) = 0$ (R_macro, Planck attractor) + micro $\lambda\_H(\Lambda\_comp) = 0$ (R_micro, doubly proven via $[T^3,T^3] = 0$ + $STr(q^4) = 0$ in ZS-S4 §6.7). Cited verbatim by ZS-A7 §4.5 Corollary V as structural license for B-2 role class.

§18.10 (mirror) — Horizon Spinor Theorem [DERIVED via ZS-A7]

PART V Chapter 18 §18.10 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis. CPTP/Choi-state PROVEN. 4π closure at Z-Spin event horizons. $\varepsilon(r\_H) = 0$ second independent path. Cross-citation locus: ZS-A7 §3.2-bis ↔ §18.10 ↔ §20.3.3 (PART VI).

§VI.2 — Limitation Catalog Row 10 [BOOK-NATIVE]

Z-Spin v1.0 corpus's catalog of known limitations. Row 10 = V1 (Planck-to-bulk handoff) matching question of ZS-M12 §7.6, the single largest open knot of Z-Spin cyclic cosmology. Cited by ZS-U11 v1.0 [40/40 PASS, capstone Paper 70] which CLOSES Row 10: V1 OPEN → DERIVED-CONDITIONAL via four independent Q-protection channels.

Chapter 5 — Gauge Couplings + Photon Synthesis + EM Duality

Spectral-to-β Bridge: Action → Polyhedral Lattice → Spectral Density → β-coefficients → Couplings. Three gaps closed in ZS-S1: '+1' in α_s via Z-sector Schur complement, x* via Berry phase, Spectral-to-β via vertex-matter + face-gauge identification. ZS-S7 (mass gap) + ZS-S12 (photon as Z-sector EM half-bridge, 5 pillars) + ZS-S15 NEW v3.0 (twin-Reuleaux EM duality, 5 pillars + Dirac-equation corollary). Pillar I: twin-Reuleaux pair (R_1, R_2) realizes (V_XZ, V_ZY) at half-angle holonomy ±θ/2; Pillar II: Poynting $S = E\times B$ equals Lie commutator $[J_{R_1}, J_{R_2}] = J_S$; Pillar III: Maxwell vacuum equations are differential form of time-point/space-point handshake duality (ZS-F8 §5); Pillar IV: SO(3) Maxwell cycle (period 2π) is Z_2 quotient of SU(2) twin-Reuleaux cycle (period 4π); Pillar V: U(1) gauge invariance = twin-Reuleaux realization of Single-Polyhedron U(1) Exactness. Corollary: Weyl spinor pair (ψ_L, ψ_R) is tensor-product realization of (V_XZ, V_ZY), with $\gamma^5 = J$. → ZS-S1, ZS-M2, ZS-M8 ($c_4 = 4/13$), ZS-M9 (McKay), ZS-S15 [35/35 PASS], ZS-M24 [35/35 PASS].

Chapter 6 — Six Forces Unified

X-Z-Y fractal symmetry: micro Weak(X)–EM(Z)–Strong(Y) ↔ macro DM(X)–Gravity(Z)–DE(Y). Cross-Coupling Theorem PROVEN (ZS-M2 §5). Axion-free Strong CP via Y-sector instanton suppression + real $m\_d/m\_u = 2e^A$ (the factor 2 origin = ZS-F12R Tetrahedral Dual Orientation Multiplicity $\mu\_Tet = 2$ NEW v3.0). Predicted nEDM $d\_n \sim 2.0 \times 10^{-41}$ e·cm. → ZS-M2 [26/26 PASS]; ZS-F12R [20/20 PASS].

Chapter 7 — Neutrinos, Hypercharge, Baryogenesis

Type-I seesaw, $m\_D = m\_e\cdot A = 40.93$ keV, $M\_R = 33.50$ GeV, $|\theta|^2 = 1.49\times10^{-12}$. Inverted Ordering canonical. μ-τ reflection ($\delta\_CP = \pm\pi/2$). $\eta\_B = (6/11)^{35}$. v2.0/v3.0: ZS-U9 Hypercharge

Trinity Braiding + automatic SM anomaly cancellation; ZS-S6 Z-Transit CP lcm(5,7) = 35; ZS-M9/M15/S11 McKay anomaly. → ZS-S2, S5, S6, U7, U9, M9.

Chapter 8 — Higgs, EW, Electron Synthesis, Master Action

ε-Higgs portal EFT, EWPT H_ZS/H_GR = 0.9622. 1-Loop Quartic Cancellation. Y-sector Spectral VEV v = 245.93 GeV. $\sigma_1/\sigma_2 = 17$ (Cabibbo principal angle), $\sigma_1/\sigma_3 = 3477$. m_τ absolute scale. Six pillars of Electron Synthesis. ZS-S10 Master Bridge U(1)_Z ↔ U(1)_Y. ~95% closure via Master Action; ZS-S14 v1.0 Master Action Total Closure 78/78 PASS, single-paper closure ≈ 99%. NEW v3.0: ZS-M20 [50/50 PASS] spectral tetrad sub-isotype assignment closes the lepton mass ratios in LOCKED corpus integers — σ_1/σ_2 = Q+Y = 17 anchored on (4-φ, 3+φ) Q-pair at (I-5)∩ρ_2 doublet (Higgs SU(2) doublet position) and X-pair (5-φ, 4+φ) splitting across (I-3) and (I-3') singletons (electron-positron CPT pair structure); $\sigma_1/\sigma_3 = (Q+Y)^2\cdot G + (Q-Z) = 3477$ EXACT. ZS-M21 [33/33 PASS] sextic invariant theory: closed family identity Σσ_i^{2k}|_{v_degen} = (Z^{2k+1}+1)/(X^{2k}·5^k) and falsification of the 1-loop Coleman-Weinberg closure for Conjecture M20.A. → ZS-S4, S8, S10, S13, S14, M11, M20, M21.

PART III — Early Universe & Cosmology

§0. Abstract

ε-field inflation with N* = 59.5 e-folds, n_s(N=60) = 0.9676 (PASS Planck), r = 0.00890 (BLOCKING discriminator vs Boyle-Finn-Turok r=0). Reheating $T_{reh} \approx 2.55\times10^{15}$ GeV. QKE-closed baryogenesis $\eta_B = (6/11)^{35}$. Z_2 mirror cosmology (PROVEN); cyclic holonomy $S_{Z_2} = 6\pi/A$; timescale hierarchy τ_n = t_P × exp(nπ/A) (DERIVED). Y-Time Dilation Theorem 5.3.1 promoted HYPOTHESIS-strong → DERIVED-CONDITIONAL strong via ZS-F10 closure of ZS-A8 §SA.7. NEW v3.0: Y-frame metric trio + cycle-index unobservability (ZS-F13 [24/24 PASS]); first-rotation closure via Stage-Torque Duality (ZS-F15R [35/35 PASS]); the five-phase cyclic cosmology now closes as a fully-derived Möbius ring.

Chapter 9 — Inflation, Reheating, ΛCDM

n_s = 0.9676 (N=60); r = 0.00890; r_ZS / r_Staro = 2.671 → ~6σ LiteBIRD discriminator. → ZS-U1 [F-X.1 BLOCKING gate]; ZS-U2 [reheating]; ZS-U4 [global fit, 53/53 PASS].

Chapter 10 — Baryogenesis & QKE

$\eta_B = (6/11)^{35} = 6.117\times10^{-10}$. Z_2 symmetry forces both sectors to generate identical asymmetry (PROVEN). Resonant ARS via Pilaftsis-Underwood. → ZS-U3, ZS-U7, ZS-S5.

Chapter 11 — Z_2 Mirror Cosmology and Cyclic Phases

$V\_E(-\varepsilon) \equiv V\_E(+\varepsilon)$ PROVEN exact. Phases A→B→C→D→E→A six-step cycle. Sequential X-clock reading (preserved) and parallel dual-clock reading (DERIVED-CONDITIONAL strong via ZS-F10 Theorem F10.2). The 10^{17} ratio $\tau_6/\tau_5 = \exp(\pi/A)$ is the X-observation of the same Y-completion event. → ZS-U8 v1.0 [60/60 PASS]; ZS-A8R; ZS-F10 [30/30 PASS]; ZS-F13 [24/24 PASS].

Chapter 12 — Global Fit and Late-Time Cosmology

Ten predictions: (H_0, Ω_m, w, S_8, Y_p, D/H, η_B, n_s, r, D_V) confronted with Planck/DES/DESI/SH0ES. ΔN_eff = 2A = 0.16018 resolves D/H from −1.8σ to −0.05σ. Ω_Λ/Ω_m = 2e^A factor 2 from ZS-F12R Tetrahedral Dual Orientation Multiplicity (NEW v3.0). → ZS-U4 [53/53 PASS], ZS-U6 (Cobaya MCMC), ZS-T1.

PART III Source Paper Map (v3.0 extended)

Source paper	Role	Light pointer
ZS-U1	ε-field inflation, F-X.1 BLOCKING gate	→ ZS-U1
ZS-U2	Reheating dynamics	→ ZS-U2
ZS-U3	Baryon asymmetry	→ ZS-U3
ZS-U4	Global cosmological fit, 53/53 PASS	→ ZS-U4
ZS-U5	Quantum Gravity Bridge, T_d symmetry, Immirzi	→ ZS-U5
ZS-U6	Cobaya MCMC + face-counting v2.0 Ω_cdm = 32/121	→ ZS-U6
ZS-U7	QKE-closed baryogenesis	→ ZS-U7
ZS-U8	Cyclic Holonomy & Z_2 Vacuum, parallel reading DERIVED-COND strong	→ ZS-U8 [60/60 PASS]
ZS-U9	Hypercharge Trinity Braiding	→ ZS-U9
ZS-U10	Pentagon Tetration & Schwinger Coefficient	→ ZS-U10
ZS-U11	Bounce Q-Survival Closure: V1 Resolution	→ ZS-U11 [40/40 PASS]
ZS-A8R (post-v2.0)	Contracting Universe Dynamics; Polyhedral-Tetration Bridges	→ ZS-A8R [22/22 PASS]
ZS-A9R (post-v2.0)	Banach–Tarski origin of doubling-halving symmetry	→ ZS-A9R [47/47 PASS, +14 v1.0(R) §11]

PART IV — Quantum Gravity Bridge

§0. Abstract

Q = 11 ↔ quantum tetrahedron (T_d symmetry). Bekenstein-Hawking ln 2 from Z_2 seam parity. Two paths to Immirzi $\gamma\_LQG = \ln 2/(\pi\sqrt{3})$. Z-Telomere RG flow, $\lambda\_vac = 2A^2$ IR-stable. Cyclic cosmology $S\_Z_2 = 6\pi/A$. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin of doubling-halving symmetry. $m\_e = \sqrt{(5 - \varphi)}$. NEW v3.0 (Math Spine RH-Bridge programme): the V_4-equivariant ZBSI framework on the BV-BFV cobordism-history fiber

(M22–M28); the four Dragons D4a–D4d of the Y-sector RH contribution map (M23, sharpened by M25–M28, integrated by M33); the Three-Wall Quantitative Map (M26 Theorem M26.3); and the spinor-cycle averaged closure of the Stapledon Bridge residual (M32) closing NC-M29.RZ-A^PK at DERIVED-CONDITIONAL.

Chapter 13 — Q = 11 Spin-Network Correspondence

$|T_d| = 24$, $|Stab\_Td(v)| = 6$, $|I_h/T_d| = 5$. $j = 1/2$ unique among half-integers giving $(Z, X, Y) = (2, 3, 6)$ (ZS-M3 Th 5.1). Explicit non-identity: $Q = 11 \neq \dim(\text{intertwiner}) = 2 \neq \dim(\text{Hilbert}) = 16$. → ZS-U5 §2-3, ZS-M3.

Chapter 14 — Z_2 Seam Dynamics

Dual κ census (κ_signflip vs κ_disrupted). Regge-Holonomy: $\delta\varphi\_cell = A$ via 7-step algebraic chain. SU(2) center pairing $D^{1/2}(-I) = -I$. → ZS-M3 [27/27 PASS]; ZS-Q4.

Chapter 15 — Micro-Macro Transition

Phase drift → tunneling action. Z-Anchor dual mechanism. $\lambda\_vac = 2A^2$ IR fixed point. Cyclic cosmology $S\_Z_2 = 6\pi/A$. Bounce Q-survival. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin. $m\_e = \sqrt{5 - \varphi}$. ZS-M17 continuum-limit theorems M17.1–M17.7 (Tiling Continuum Convergence, Lieb-Robinson Tightness, Reflection Positivity, Path-Integral Closure of ΔΓ_G2, Pre-Geometric Metric Emergence, Continuum Limit Universality, Wightman QFT Reconstruction). → ZS-U5, U8, U11 (closes V1), A6 §4.5.6 (F-A6.1 CLOSED), A8R, A9R, M12, M14, M17.

Chapter 15A — NEW v3.0 — The Math Spine Riemann-Hypothesis Programme (M22–M33)

Cumulative 47+ exploratory rounds consolidated by ZS-M22 v1.1 (Five-Pillar Arithmetic-Dedekind Scaffold) into a non-RH-claiming structural map of where Z-Spin and Riemann ride together (dong-seung). The programme's logical structure:

- ZS-M22 v1.1 [60/60 PASS] — Five Pillars: I (n=3 Eisenstein chain), II (Q=11 cyclotomic chain), III (multiplicative arithmetic fiber), IV (critical line as J-symmetry locus), V (scalar-kernel no-go for Weil positivity). ADS-6 PROVEN no-go on boundary fiber forces ADS-H1 cobordism BRST positivity hypothesis as the sole surviving route.
- ZS-M23 v1.0(Revised) [31/31 PASS] — Y-Sector RH Contribution Map. Four Dragons D4a (V_4-Sonin embedding), D4b (Burnol conductor), D4c (defect-square realization), D4d (cobordism BRST-Hodge projection). NC-M23.7: D4 closure alone does not close GRH-for-K.
- ZS-M24 v1.0 [35/35 PASS] — Face polygon spectral zeta $\zeta^*\nabla^*(s)$. Theorem D.1 Legendre duplication archimedean decomposition. Theorem D.2 B(s) identification. P2 status OPEN → PARTIAL.
- ZS-M25 v1.0 [26/26 PASS] — Composite biquadratic $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. Theorem D.1-K four-factor archimedean decomposition $\xi\_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(s, \chi\_{-3}) \cdot \Lambda(s, \chi\_{-11}) \cdot \Lambda(s, \chi\_{33})$. Theorem P3-J J-Yakaboylu compatibility (J commutes with H_Q^Yak,J on $\sigma = 1/2$).

• ZS-M26 v1.0 [24/24 PASS] — V_4-Equivariant ZBSI on cobordism-history fiber. Theorem M26.1 V_4-Schur idempotent decomposition; Theorem M26.2 Projected Determinant per Channel (C_χ ∈ {1, √3, √11, √33}); Theorem M26.3 Three-Wall Quantitative Map (W1, W2, W3).
• ZS-M27 v1.0 [24/24 PASS] — V_4-Equivariant Cobordism BRST Closure via Kostant Cubic Dirac. Closes Wall W3 at DERIVED-CONDITIONAL via three external imports (Alekseev–Barmaz–Mnev 2018, Huang–Pandžić 2002, Kostant 1999/2003).
• ZS-M28 v1.0 [30/30 PASS] — The Z-Spin RH Bridge. Theorem 28.5 σ = 1/2 ↔ j = 1/2 isomorphism (DERIVED-interpretation). Corollary 28.5a Möbius-trace infinity (HYPOTHESIS-strong). Theorem 28.14 Y-sector geometric carrier of χ_−3 (DERIVED-CANDIDATE).
• ZS-M29 v1.0 [79/79 PASS] — Z-Funnel Spectral Retraction with Stapledon Bridge. Theorem 9.1 conditional bridge from string compactification (A_5-Hilb(X_F) Calabi–Yau threefold, h^{1,1} = 5, h^{2,1} = 15) to the Z-Spin truncated-icosahedron Hodge–Dirac sector. NC-M29.RZ principal OPEN gate.
• ZS-M30 v1.0 [32/32 PASS] — Z-Spin Duality and the RH Bridge. Theorem 30.1 Möbius-Trace Infinity (DERIVED-interpretation strong, six-route over-determination). Theorem 30.2 Y-Sector Two-Face Spectral Mirroring. Theorem 30.5 Twenty-One 1/2 Catalogue across six categories. Three-layer duality: Sector / Frame / Protocol.
• ZS-M31 v3.0 [36/36 PASS] — W2 as Cross-Coupled Sector Duality. Lemma M31.0 Non-Separability (no sum-form W_K = F_X + F_Y + F_Z compatible with corpus data, max variance 13.011). Theorem M31.4 Z_2-Parity Selection Rule. Self-Dual Replication Principle (SDRP) registered.
• ZS-M32 v1.0 [56/56 PASS] — Spinor-Cycle Averaged Residual Criterion. Path-Reversal Lemma M32.X (B^PK preserves phase sign, contrasting B^† which flips it). Two-layer phase quantum α_amp = π/10 (amplitude) / α_op = π/5 (operator). N = 20 spinor-identity closure: R_Z^ZS|_{Z-even}^{4π-avg} = 0 EXACTLY at N = 20. Closes NC-M29.RZ-A^PK at DERIVED-CONDITIONAL.
• ZS-M33 v1.0 [52/52 PASS] — V_4-Equivariant Weil Functional Closure via Path γ-revised Z-Mediator Reading. Reading C (NEW): all four V_4-arithmetic mechanisms route through Z-mediator projector Π_Z = (1/2)(I + J_Z). Sub-targets D4a (CCM 2024), D4b (Burnol 1998 + log additivity log(3)+log(11)=log(33)), D4c (Wilson-LOCATOR phase factor), D4d (Kostant cubic Dirac BRST cohomology) integrated into single Tr identity. Sign-flip mechanism decomposition (5/12 NEG → predicted 12/12 POS).
Programme NON-CLAIM (NC-M23.1 preserved verbatim throughout): Z-Spin does not, and structurally cannot, prove RH from internal data alone. The walls W1, W2, W3 are mapped, sharpened, and progressively closed; the program's contribution is a coordinated three-axis over-determination of the Bridge thesis, not a proof.

PART IV Source Paper Map (v3.0 extended)

Source paper	Role	Light pointer
ZS-U5	Quantum Gravity Bridge, T_d symmetry, Immirzi	→ ZS-U5
ZS-U8	Cyclic Parallel Reading	→ ZS-U8 [60/60 PASS]

ZS-U11	Capstone: V1 Planck-bulk CLOSED	→ ZS-U11 [40/40 PASS]
ZS-M3	Regge-Holonomy + Immirzi + Z-Telomere	→ ZS-M3 [27/27 PASS]
ZS-M12	Auto-Surgery, $\varepsilon_{min} \approx 30.7$	→ ZS-M12
ZS-M14	Geometric $m_e = \sqrt{5-\varphi}$	→ ZS-M14 [47/47 PASS]
ZS-M17	Continuum limit, OS reconstruction	→ ZS-M17 [60/60 PASS]
ZS-A6	Z-Anchor + cigar bounce, F-A6.1 CLOSED	→ ZS-A6 §4.5.6 [140/140]
ZS-A8R	Polyhedral-Tetration Bridges	→ ZS-A8R [22/22 PASS]
ZS-A9R	Banach-Tarski origin	→ ZS-A9R [47/47 PASS]
ZS-F13	Möbius chronology, ring closure	→ ZS-F13 [24/24 PASS]
ZS-M18 (post-v2.0)	Free-Exploration Log: Speculative Prime-Polyhedral Correspondences	→ ZS-M18 [51/51 PASS]
ZS-M22 v1.1 (post-v2.0)	Five-Pillar Arithmetic-Dedekind Scaffold	→ ZS-M22 [60/60 PASS]
ZS-M23 v1.0(R)	Y-Sector RH Contribution Map; four Dragons D4a–D4d	→ ZS-M23 [31/31 PASS]
ZS-M24	Face polygon spectral zeta + Archimedean Completion	→ ZS-M24 [35/35 PASS]
ZS-M25	Composite-Field $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ + J-Yakaboylu Bridge	→ ZS-M25 [26/26 PASS]
ZS-M26	V_4-Equivariant ZBSI on Cobordism-History Fiber	→ ZS-M26 [24/24 PASS]
ZS-M27	Cobordism BRST Closure via Kostant Cubic Dirac (W3)	→ ZS-M27 [24/24 PASS]
ZS-M28	The Z-Spin RH Bridge	→ ZS-M28 [30/30 PASS]
ZS-M29	Z-Funnel Spectral Retraction + Stapledon Bridge	→ ZS-M29 [79/79 PASS]
ZS-M30	Z-Spin Duality and the RH Bridge — 21 halves	→ ZS-M30 [32/32 PASS]
ZS-M31 v3.0	W2 as Cross-Coupled Sector Duality	→ ZS-M31 [36/36 PASS]
ZS-M32	Spinor-Cycle Averaged Feshbach Residual; closes NC-M29.RZ-A^PK	→ ZS-M32 [56/56 PASS]
ZS-M33	Path γ-revised V_4-Equivariant Weil Functional Closure	→ ZS-M33 [52/52 PASS]

PART V — Astrophysics & Strong-Field

§0. Abstract

Galaxy rotation without dark matter (ε-Halo, $r_Z = 18.3$ kpc). Neutron stars +3.93% mass, tidal $\tilde{\Lambda}$ +21.24%. Black holes: Wald entropy, Z-instanton, sector duality. Information paradox via seam witness; signed witness $\tilde{u}_{seam}$ (F-A7.3 = cheapest near-term gate). 991/991 PASS cumulative across PARTS V+VI+VII preserved. NEW v3.0: ZS-A8 v1.0(Revised) [22/22 PASS] — contracting-universe dynamics with Polyhedral-Tetration Bridges 1–3 and Symmetry–Asymmetry Unified View; ZS-A9 v1.0(Revised) [47/47 PASS] — Banach–Tarski origin of cosmological doubling-halving symmetry through the $F_2 \to D_4$ amenability functor and the Julia set $J(T)$ of $T(z) = i^z$ as a measure-theoretic boundary.

Chapter 16 — Galaxy Rotation Without Dark Matter

ε-Halo from Goldstone mode, terminus r_Z = 18.3 kpc. Modified gravity $\mu = \Sigma = 1/(1+A)$. Three convergent routes $\Omega_m = 38/121$. Factor 2 in $\Omega_\Lambda/\Omega_m = 2e^A$ from ZS-F12R (NEW v3.0: $\mu_{Tet} = 2$ from Tetrahedral Dual Orientation Multiplicity, F-Tet-2eA-1 REJECTED with replacement F-Tet-2eA-2). → ZS-A1 [78/78 PASS]; ZS-A5 [50/50 PASS]; ZS-S3 [25/25 PASS]; ZS-F12R [20/20 PASS].

Chapter 17 — Neutron Stars, GW, LSS

NS M_{max} +3.93% uniform across EOS (universal $G_{eff} = G/(1+A)$). Tidal $\tilde{\Lambda}$ +21.24%. → ZS-A2.

Chapter 18 — Black Holes (incl. §18.10 Horizon Spinor)

ε-field horizon, Wald entropy, Z-instanton + GW signatures. $\varepsilon(r_H) = 0$ PROVEN (cigar bounce). §18.10 v2.0 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis [PROVEN, CPTP/Choi]. → ZS-A3 [49/49]; ZS-A6 [140/140]; ZS-A7R [38/38].

Chapter 19 — Information Paradox Resolved

Seam witness u_{seam}, signed witness $\tilde{u}_{seam}$, Q = 11 register, lattice gauge P1. F-A7.3 = 2π vs 4π discrimination on hardware ~2028-2032 (cheapest gate, ~8% incremental shot cost). → ZS-A4 [54/54]; ZS-A7R §C [38/38].

Chapter 20A — NEW v3.0 — Contracting Universe Dynamics (ZS-A8R)

ZS-A8 v1.0(Revised) [22/22 PASS] introduces the SYMMETRIC contraction dynamics: the wave-to-particle transition controlled by the dual factor (1−2A). Three Bridge formulas: Bridge 1 [OBSERVATION-strong] $|z^*| \approx B = \delta_X + \delta_Y$ at 0.0087%; Bridge 2 [OBSERVATION-strong] $\eta_{topo} \approx B^2 + (\delta_Y - \delta_X)^2/[Y^2\cdot(1-2A)]$ at 0.000001% ($Y^2 = 36 = E(TO)$ wave-channel scale); Bridge 3 [DERIVED-CONDITIONAL via slog-L2 equivalence] $\Delta = 1/2 - x^*$ via L2 self-locking iteration at 0.000002%. Y-Time Dilation Theorem 5.3.1 (DERIVED-CONDITIONAL strong via ZS-F10): X-clock observation of Y-completion is dilated by $\exp(\pi/A) \approx 10^{17}$ per added Y-dimension. $\exp(\pi/A) = \exp(N_{(2\pi)} \times \langle\sin^2(\varphi/2)\rangle) = \exp((2\pi/A) \times (1/2))$. Symmetry-Asymmetry Unified View (§SA): the universe tilts simultaneously by (1+A) (X-Inward expansion) and (1−2A) (Y-Outward contraction); only the impedance A = 35/437 is frame-invariant.

Chapter 20B — NEW v3.0 — Banach–Tarski Origin of Cosmological Doubling-Halving Symmetry (ZS-A9R)

ZS-A9 v1.0(Revised) [47/47 PASS, 12 OPEN closures] establishes that the (1+A) ↔ (1−2A) duality of late-time cosmology is the measure-preserving quantization of the Banach–Tarski paradoxical decomposition through the i-tetration fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$. Three theorems: ZS-A9.1 (DERIVED) $F_2 \to D_4$ amenability functor — the BT engine $F_2 \subset SO(3)$ maps to the amenable register dihedral $D_4 = \langle J, J_Z\rangle$, converting BT non-amenability to Z-Spin amenability; ZS-A9.2 (HYPOTHESIS-strong) the Julia set J(T) of T(z) = i^z is a measure-zero, totally disconnected, cardinality-$2^{\aleph_0}$ set serving as a constructive ZF-analogue of the BT non-measurable set; ZS-A9.3 (DERIVED-CONDITIONAL) two-branch decomposition $(1+A)(1-2A) = 1 - A(1+2A)$ with X-Inward macroscopic expansion exp(A), Y-Outward microscopic contraction $Y^2(1-2A)$, and Z-mediator absorbing the residual $A(1+2A) \approx 9.29\%$ via $\kappa^2 = A/Q = 35/4807$. v1.0(Revised) §11 closes all four OPEN items via the *-homomorphism reformulation, the Świerczkowski geometric realization of free F_2-rotations, and the inheritance of dx_X + dx_Y + dx_Z = 0 from ZS-Q7 §5 master-equation probability conservation. OPEN-2.B (formal ZF/ZFC+AC isomorphism between BT non-measurable and J(T)) recategorized as PERMANENT NC under Solovay 1970.

PART V Source Paper Interface (v3.0 extended)

Source paper	Verification	Primary chapter	Key result
ZS-A1	78/78 PASS	Ch 16	Rotation curves, BTFR, M-σ, Keplerian
ZS-A2	verified	Ch 17	NS M_max, $\tilde{\Lambda}$, GW c_T = c
ZS-A3	49/49 PASS	Ch 18 §1-7	ε-field, Wald, Z-instanton
ZS-A4	54/54 PASS	Ch 19	Seam witness, lattice P1
ZS-A5	50/50 PASS	Ch 16 §16.7	Three convergent Ω_m = 38/121
ZS-A6	140/140 PASS	Ch 18 §18.2 + §18.8	Cigar bounce, F-A6.1 CLOSED
ZS-A7R	38/38 PASS	Ch 18 §18.10 + Ch 19 §19.7-8	Horizon Spinor 4π
ZS-A8R (post-v2.0)	22/22 PASS	Ch 20A	Polyhedral-Tetration Bridges 1–3, Symmetry-Asymmetry View
ZS-A9R (post-v2.0)	47/47 PASS	Ch 20B	Banach-Tarski origin of doubling-halving symmetry
ZS-S3	25/25 PASS	Ch 16 §16.8 + Ch 17 §17.3	Modified gravity
ZS-F12R	20/20 PASS	Ch 16 §16.3	Layer Separation factor 2 origin (μ_Tet)

PART VI — Quantum Mechanics

§0. Abstract

Densest PART [376/376 PASS]. Geometric decoherence τ_D/τ_Penrose = 1/A = 12.49. Bell CHSH = $2\sqrt{2}$ from dim(Z) = 2. Proton spin Hodge ½ΔΣ = 3/22, ΔG = 2/11. Lattice gauge a_2 = 19/6 (TO 0.08σ Wilson MC). CP Jarlskog from δφ = A. Z-Spin Quantum Architecture (QH/QS/QC). Inverse Riemann Engine. Structural Arrow of Time (ZS-Q7). Cross-link with v3.0 RH-Bridge programme (M22–M33) at the ZBSI cobordism-history fiber level.

Chapter 20 — Geometric Decoherence + Born Rule

Stinespring + Lindblad $\Gamma = 2A(\Delta E/\hbar)^2$. $\tau_D = 12.49$. Born rule DERIVED from $L_{XY} \equiv 0$. NC-F11.1–6 inherited verbatim §20.7. → ZS-Q1 [29/29 PASS].

Chapter 21 — Bell, Entanglement, Holographic Conjecture

CHSH = $2\sqrt{2}$, $\tau_{ent} = \tau/2$. Area Law $\chi = 2$, $\varepsilon_{BM} \to 0$. → ZS-Q2 [48/48]; ZS-Q6 [42/42].

Chapter 22 — Proton Spin from Polyhedral Hodge

BCC T^3 Hodge spectrum, $\frac{1}{2}\Delta\Sigma = 3/22$, $\Delta G = 2/11$. $(V+F)_X = 38$ Mode-Count Collapse. → ZS-Q3 [40/40].

Chapter 23 — Lattice Gauge QSim on TO

$a_2 = (V+F)/G = 19/6$ PROVEN topological. Wilson MC TO 0.08σ. J-parity. → ZS-Q4.

Chapter 24 — CP Violation and UV Limits

Jarlskog from $\delta\varphi = A$. UV sufficiency of polyhedral spectra. → ZS-Q5.

Chapter 25 — Bond Dimension, Holographic Area Law, Markov Limit

Bond dimension $\chi = 2 = \dim(Z)$; Regge area law derived. $\varepsilon_{BM} = 2/Q$ Markov limit purely geometric. → ZS-Q6 [42/42 PASS].

Chapter 26 — Structural Arrow of Time

$\Gamma(X\to Y)/\Gamma(Y\to X) = \dim(Y)/\dim(X) = 2$ PROVEN; $\Delta S = \ln 2$ per Z-transit. Coarse-grained entropy bias from $L_{XY} = 0 + \dim(X) \neq \dim(Y)$; microscopic dynamics remain time-reversal symmetric. → ZS-Q7 [40/40 PASS].

PART VI Source Paper Interface

Source paper	Verification	Primary chapter	Key result
ZS-Q1	29/29 PASS	Ch 20	Geometric decoherence, Born rule
ZS-Q2	48/48 PASS	Ch 21	Bell, entanglement, holographic conj.
ZS-Q3	40/40 PASS	Ch 22	Proton spin Hodge
ZS-Q4	verified	Ch 23	Lattice gauge TO
ZS-Q5	verified	Ch 24	CP violation, UV limits
ZS-Q6	42/42 PASS	Ch 25	Area law, Markov limit
ZS-Q7	40/40 PASS	Ch 26	Structural arrow of time
ZS-QH	42/42 PASS	Ch 25/29	Quantum hardware FMD specifications
ZS-QS v1.0(R)	35/35 PASS	Ch 24/29	Inverse Riemann Engine: Triple Structure (DETECTOR + LOCATOR + EXCLUDER) and Boolean Resonance Filter

ZS-QC	52/52 PASS	Ch 29	Quantum architecture system integration

PART VII — Translational and Computational

§0. Abstract

$\Delta N\_eff = dim(Z) \cdot A = 2A = 0.16018$ → BBN D/H −1.8σ → −0.05σ (ZS-T1). Anti-numerology MC: 5670 × 29 = 164,430 candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS — framework MORE constrained than random). Tier-1 ★★★ T1-1: $1-n\_s = A \cdot x^*$ (0.008%, 0.00σ vs Planck). T1-2: $\alpha\_EM \approx A/Q$ (0.22% LO, 0.00007% NLO Schur) (ZS-T2). 8/8 closure parameters DERIVED via three chains; $n\_s(N^* = 55) = 0.9676$; $r(N^* = 60) = 0.0091$ (ZS-T3). NEW v3.0: ZS-T4 v1.0 [36/36 PASS] — Cosmos-Human Isomorphism: six-step telomere hierarchy and (Body, DNA, Brain) sector decomposition as a two-scale realization of the Z-Spin block architecture; structural, substrate-agnostic, NON-CLAIM that humans physically realize Z-Spin geometry.

Chapter 27 — Phenomenological

$\Delta N\_eff = dim(Z) \cdot A = 2A = 0.16018$ → BBN D/H −1.8σ → −0.05σ. Block Fiedler Mediation Theorem PROVEN (T1-2/T1-3 reciprocal duality $\lambda_2 = 2A/Q$ with X-face propagator scale $1/\kappa^2 = Q/A \approx 137$ and Y-face vertex coupling $\kappa^2 = A/Q \approx 0.0073$). → ZS-T1 [42/42].

Chapter 28 — Spectral Observatory

Anti-numerology MC: 164,430 candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS). Tier-1 ★★★ T1-1: $1-n\_s = A \cdot x^*$. T1-2: $\alpha\_EM \approx A/Q$ (0.00007% NLO via McKay $c_4 = 4/13$). → ZS-T2 [30/30].

Chapter 29 — Z-Sim Forward Simulator + Quantum Architecture

8/8 closure parameters DERIVED via three chains: ZS-Q7 §5.8 + ZS-M3 §10 + ZS-Q5 §5.2. Eigenvalue $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ to 10^{-17}. 500K MC universal-theorem confirmation. $n\_s(N^* = 55) = 0.9676$ PASS Planck; $r(N^* = 60) = 0.0091$ PASS BK18. ZS-QH v1.0 functional material specifications (X = bilayer graphene QD, Z = Bi_2Te_3 topological insulator, Y = C_{60}/CNT phonon reservoir as primary candidate stack); ZS-QS v1.0(Revised) Triple Structure Theorem (DETECTOR confirmed, LOCATOR via argmax, EXCLUDER via argmin) and Boolean Resonance Filter (XOR identity with ZS-F8 (E, R) handshake operators); ZS-QC v1.0 system integration. → ZS-T3 [35/35]; ZS-QH; ZS-QS v1.0(R); ZS-QC [52/52].

Chapter 30 — NEW v3.0 — The Cosmos-Human Isomorphism (ZS-T4)

ZS-T4 v1.0 [36/36 PASS] establishes a structural isomorphism between Z-Spin cosmology and the human organism at two parallel levels, with Cardinal NC-4 (Z-Brain) explicitly extended to body and genome — NO physical realization claim. Level I: cosmological timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$, $n \in \{0,...,6\}$, mapped onto a six-step human life-cycle (gametic union → cleavage → stem-cell expansion → somatic growth → reproductive plateau → senescence → death/inheritance). Strong anchors at n=2 (mediator coset Z=2 ↔ stem cells, HYPOTHESIS-strong), n=5 (proton decay ↔ senescence, HYPOTHESIS-strong), n=6 (Z_2 holonomy ↔ death/inheritance, HYPOTHESIS-strong). Level II: (X, Z, Y) = (3, 2, 6) sector decomposition mapped onto (Body, DNA, Brain) tripartite anatomy at organism scale. Inherits ZB-P7 v1.0 cosmic-biological VERIFIED status (ΔPCI(Wake−Anesthesia) ≈ η_topo at 0.7% empirical / 1.5% in-silico). Hayflick limit ≈ Z-Telomere completion count $N_{(2\pi)} = 2\pi/A \approx 78.45$ registered as HYPOTHESIS / TESTABLE-LONG with 5-year derivation-or-retraction horizon. Twelve falsification gates F-T4.1–F-T4.12 pre-registered.

PART VII Source Paper Interface (v3.0 extended)

Source paper	Theme/Role	Chapter	Verification
ZS-T1	Phenomenological	Ch 27	42/42 PASS
ZS-T2	Observational	Ch 28	30/30 PASS
ZS-T3	Computational (Z-Sim)	Ch 29	35/35 PASS
ZS-T4 (post-v2.0)	Cosmos-Human Isomorphism	Ch 30	36/36 PASS
ZS-F1, F2, F5	Foundations cross-link	all	PROVEN
ZS-M1, M3, M6, M8, M11	Math Spine cross-link	various §	verified
ZS-Q1, Q5, Q7	QM cross-link (chains)	Ch 29 §29.1-4	verified
ZS-U1, U6	Early Universe cross-link	Ch 28 §28.3, Ch 29	verified
ZS-QH/QS/QC	Hardware/Software/Integration translational	Ch 29	42 + 35 + 52 PASS

§VII.B-LIGHT — Seven Critical Near-Term ★ Gates (Timeline-Anchored)

These seven gates will either PASS or FALSIFY Z-Spin within ~10 years. Source: §V.G.6, §VI.G.11, §VII.B. Preserved verbatim from v2.0 Light.

#	Gate ID	Test	Timeline	Status
1	F-A1.14 ★	Gaia DR4 MW outer rotation: Keplerian decline beyond 30 kpc (ε-Halo finite vs particle NFW)	~2026	Earliest decisive
2	F-A1.15 ★	Gaia DR4/DR5: r_Z(MW) outside [10, 50] kpc (Z-Spin = 18.3 kpc)	~2026-2030	Pending
3	F-A2.5 ★	DESI/Euclid/Rubin: S_8 outside [0.75, 0.84] (~1% precision)	~2026-2030	Pending
4	F-A7.3 ★	Z-Spin hardware: 2π vs 4π discrimination on signed seam witness ũ_seam	~2028-2032	Cheapest (~8% shot overhead)

5	F-X.1 (LiteBIRD) ★	r outside Z-Spin band 0.0085-0.0094 (BLOCKING; ~9σ vs Boyle-Finn-Turok r=0)	~JFY2032	BLOCKING
6	F-A3.7 ★	Hyper-K: τ_p outside [10^33.5, 10^35.5] yr (Z-Spin = 2.56×10^34)	~2030-2035	Pending
7	F-A2.1, F-A2.2 ★	10+ pulsars: NS M_max +3.93% uniform; ET/CE: Λ̃ +21.24% to ~5%	~2030s, ~2035	Pending

PART VIII — Methodological Toolkit: Solution Tools for Stuck Equations

Purpose. This Part is the operational core of v3.1 Light. It promotes the v3.0 Light from a structural pointer-index ("where to read") to a working solution kit ("how to proceed when stuck"). Each tool below is distilled from a corpus paper that achieved a methodological breakthrough — typically by reframing rather than by adding new mathematics. When an AI agent or human researcher encounters a stuck equation, an OPEN flag, or an apparent contradiction inside the Z-Spin framework, this Part is the first place to consult.

Reading order. PART VIII is organised as nine numbered tools (T1–T9). Each tool is given in a fixed schema: (a) Trigger — the symptom that suggests this tool; (b) Source — the originating paper and §-anchor; (c) Statement — the operational kernel in compressed form; (d) Worked corpus precedent — the OPEN flag the tool was first used to close; (e) Generalisation — the broader class of stuck equations the tool can be re-applied to. PART IX (Insight Gallery) extends the tool kit with the deeper conceptual reframings (sector-equivalences, frame-transformations, cross-coupling), and the Master Bypass Index (after PART IX) provides the symptom → tool lookup table.

§VIII.1 — Tool T1: Five-Axiom Diagnostic Lattice (F18)

Trigger.

(a) An equation appears stuck and the failure mode is unclear; (b) a derivation looks circular; (c) several papers address "the same question" but their results don't compose. In any of these cases, T1 isolates which of the five structural axioms is failing.

Source.

ZS-F18 v2.1 §3 + §6.6 (Table 6.1) + §7.4. The five axioms $\mathscr{A}$ = {Existence, Self-reference, Algebra, Non-triviality, Unitarity} and the Four Bridges from $\mathscr{A}$ to the operational corpus.

Statement (operational form).

When stuck, ask which axiom is the failure attributable to. Each axiom has a Bridge (theorem) into the corpus; locate the Bridge and ask whether the bridge is intact for the present problem. The five-axiom set is exhaustive for Z-Spin physical content (ZS-F18 §5, INTERPRETATION-strong); a stuck equation that does not map to any of the five is either a numerology candidate (Anti-Numerology Gate, ZS-F2) or a category error (frame mismatch — see Tool T6).

Axiom	Failure symptom	Controlling Bridge	First port of call

A1 Existence	Object referenced but not constructed	FFPP nested compression	ZS-F0 §13 Theorem 13.3
A2 Self-reference	Definition uses what it defines (no fixed-point)	Bridge One (Cross-Coupling)	ZS-M2 §5 PROVEN; ZS-M1 i-tetration
A3 Algebra	Numerical balance fails / numerology	Master Equation (algebraic)	ZS-M1 §4 + ZS-F17 §3
A4 Non-triviality (algebraic)	Two-channel interface collapsed	Bridge Two (NOT-AND handshake)	ZS-F8 §4 PROVEN
A4 Non-triviality (geometric)	Dimension dim(X)=3 not derived	Bridge Four (Commutator Generator)	ZS-F18 §6 + ZS-S15 §3
A5 Unitarity	Probability/information not preserved	Bridge Three (Z-Bottleneck)	ZS-Q7 Thm 2 + ZS-F8 §4
Sixth polarity	Discrete vs continuous mismatch	X-Y Tiling Asymmetry	ZS-M6 §5.5 PROVEN; ZS-M17

Worked corpus precedent.

F18 itself diagnosed 12 of the corpus's most consequential identifications (ZS-F18 Appendix A) by mapping each to its responsible axiom. The diagnosis produced no new theorem yet revealed why each result is robust — a model usage of T1 as a methodological tool.

Generalisation.

Any future Z-Spin paper encountering an OPEN flag should run T1 first. If the axiom is identified, the closure path is the corresponding Bridge; if no axiom matches, the OPEN is either out-of-scope (NON-CLAIM territory, Tool T7) or a category error (Tool T6, frame mismatch).

§VIII.2 — Tool T2: Dimensional Reduction via Bridge Four (F18)

Trigger.

Stuck on a high-dimensional construction (dim 3, 6, or 11) where the dependency chain becomes intractable. T2 reduces the problem to dim(Z) = 2 — the only foundational dimension — and lets dim(X)=3 and dim(Y)=6 re-emerge by Lie commutator and product respectively.

Source.

ZS-F18 §6.4–§6.7 (Bridge Four, dimensional origin table). ZS-S15 §3 Theorem S15.2 (Poynting–Commutator Theorem, DERIVED).

Statement.

Two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair (V_XZ, V_ZY), composed at 90° NOT-AND phase, generate a third axis $J\_S = [J\_\{R_1\}, J\_\{R_2\}]$ by Lie commutator. Therefore (Z, X, Y) = (2, 3, 6) is generated from dim(Z) = 2 alone; dim(Y) = dim(X)·dim(Z) = 6; Q = Z + X + Y = 11.

Worked corpus precedent.

ZS-F18 v2.0 used T2 to read the question "why does physical space have dimension three?" not as an additional axiom but as a derived consequence of dim(Z) = 2 plus 90° NOT-AND composition. This converted what would have been a 13th independent encounter into the geometric face of axiom A4.

Generalisation.

When facing a stuck calculation in 3D or 6D, examine whether the structure can be re-expressed as two dim(Z) = 2 channels meeting at orthogonal phase. If yes, the third axis is automatic. Application targets include any future SO(3) or SU(2)-related symmetry breaking in PARTS II–V where dimensional emergence is at stake.

§VIII.3 — Tool T3: Hermitian-to-Path-Reversal Switch (M32)

Trigger.

A Feshbach-type residual or sandwich expression refuses to vanish under the standard Hilbert-space adjoint reading. The phase factors cancel "too cleanly" or "not at all", giving either a trivial-positive or unbounded-positive result that doesn't engage with the corpus structure.

Source.

ZS-M32 v1.0 §2 (R_Z^H vs R_Z^{ZS} separation), §3 (Path-Reversal Lemma M32.X). Underpinned by ZS-S6 §4.1 $K_{bwd} \neq K_{fwd}^\dagger$ PROVEN at kernel level and ZS-T1 §10.5 PK-Conjugation Theorem T9 DERIVED.

Statement (Path-Reversal Lemma M32.X).

Under the corpus PK-Conjugation T9 plus the Regge T-odd scalar mechanism, $B \propto e^{+ik\alpha} \Rightarrow B^{PK} \propto e^{+ik\alpha}$, in contrast with the standard adjoint $(e^{+ik\alpha} \cdot B)^\dagger = e^{-ik\alpha} \cdot B^\dagger$. The path-reversal residual $R_Z^{ZS} = B \cdot Q^{-1} \cdot B^{PK}$ is a structurally distinct object from the Hermitian residual $R_Z^H = B \cdot Q^{-1} \cdot B^\dagger$; averaging-and-vanishing claims hold for R_Z^{ZS}, not R_Z^H.

Worked corpus precedent.

Closed NC-M29.RZ-A^PK at DERIVED-CONDITIONAL: $R_Z^{ZS}|_{Z\text{-even}}^{4\pi\text{-avg}} = 0$ EXACTLY at $N = 20$. The Stapledon Bridge (ZS-M29 Theorem 9.1) was upgraded from DERIVED-CONDITIONAL to DERIVED in the spinor-cycle averaged sense without any external numerical input. The closure was achieved purely by switching the operator-reading from R_Z^H to R_Z^{ZS}.

Generalisation.

Any sandwich expression in the framework that fails to close under Hermitian reading should be re-examined under PK reading. Candidate targets: future Feshbach-like reductions in PARTS IV–VI; any operator residual that involves K_{bwd} (T-odd) on one side and K_{fwd} on the other.

§VIII.4 — Tool T4: Two-Layer Phase Quantum Decomposition (M32)

Trigger.

Two corpus values that should be equal differ by exactly a factor of 2; or, an unexpected factor 2 appears that cannot be attributed to SO(3)/SU(2) double-cover. T4 separates amplitude-layer quanta from operator-layer quanta.

Source.

ZS-M32 §4 (two-layer phase quantum). Independent corpus inputs: ZS-S6 §G.2 PROVEN (vertex Regge deficit $\delta_X^{vertex} - \delta_Y^{vertex} = \pi/6 - \pi/15 = \pi/10$) and ZS-S6 §G.4 reading ii PROVEN (Z-quarter-turn Z_5 partition $(1/4)(2\pi/5) = \pi/10$).

Statement.

Z-Spin operator-level phase quantum $\alpha_{op} = \pi/5$ decomposes as $\alpha_{op} = 2 \cdot \alpha_{amp}$ where $\alpha_{amp} = \pi/10$ is the amplitude-level quantum. The factor 2 between layers is forced by the Path-Reversal Lemma applied to the Feshbach sandwich; it is the operator-layer analogue of the SO(3)/SU(2) double-cover (ZS-S15 §5).

Worked corpus precedent.

Reconciled four PROVEN occurrences of $\pi/10$ (vertex Regge, Z-quarter-turn) with PROVEN occurrences of $\pi/5$ (commutator $[\sigma_3, \hat{n}\_Y] = -2i\cdot\sin(\pi/5)\cdot\sigma_2$ in ZS-S6 §3.3, BCH leading order in ZS-S6 §3.4) under a single layer-doubling identity. Before T4, these were treated as independent constants.

Generalisation.

When two PROVEN corpus quantities differ by factor 2 with no obvious double-cover origin, check whether one is an amplitude-layer quantum and the other an operator-layer quantum. The factor 2 is then the structural signature of the layer transition, not a free parameter or coincidence.

§VIII.5 — Tool T5: Spinor-Cycle Averaging at N = 20 (M32)

Trigger.

A residual or sum that does not vanish at single iterate, but whose single-iterate phase is a primitive 10th root of unity. The standard algebraic minimum N = 10 produces zero algebraically but leaves the spinor in fermion-sign-flipped state D^{1/2}(2π) = −I, breaking physical closure.

Source.

ZS-M32 §5 (Theorem M32.3, spinor-cycle averaged vanishing). ZS-M3 Lemma 10.1 PROVEN (D^{1/2}(4π) = +I). ZS-S15 §5 (SO(3)/SU(2) double-cover ratio).

Statement.

When R_Z^ZS,(k) ∝ exp(ikπ/5)·R_Z^ZS,(0), the geometric series Σ_{k=0}^{19} exp(ikπ/5) = 0 EXACTLY, but only at N = 20 (not at N = 10) does the cumulative spinor phase reach 4π, restoring D^{1/2}(4π) = +I. The minimum physical-closure averaging window is N = 20.

Worked corpus precedent.

R_Z^ZS|_{Z-even}^{4π-avg} = 0 EXACTLY at N = 20. ZS-M32 Table 7 explicitly compares N ∈ {10, 20, 30, 40} and identifies the minimum physical closure as N = 20. The averaging structure aligns with three other corpus 4π averages: ⟨sin²(φ/2)⟩_{[0,4π]} = 1/2 (ZS-M3), ⟨V_XZ·V_ZY⟩ = 1 (ZS-T1), ⟨ε_+ + ε_−⟩ = 0 (ZS-U1) — Table 8 of ZS-M32.

Generalisation.

Any non-vanishing single-iterate quantity whose phase is a primitive 10th root of unity admits closure under N = 20 spinor-cycle averaging. Candidate targets: future Feshbach-like reductions in PARTS IV/VI; any operator-residual whose phase quantum is α_op = π/5.

§VIII.6 — Tool T6: Three-Way Split of OPEN Flags (M32)

Trigger.

An OPEN flag too large to close in one step. Attempting full closure produces circularity or external-import-blocking. T6 splits the OPEN into three structurally distinct sub-claims with different status levels.

Source.

ZS-M32 §6 (NC-M29.RZ three-way split). General pattern adoption from ZS-M22, ZS-M23R Four Dragons D4a–D4d.

Statement.

An OPEN flag NC-X may be split as: A^PK = the part closeable under path-reversal/averaging now (DERIVED-CONDITIONAL); B = the part with structural evidence but one identified gap

(HYPOTHESIS-strong, with explicit closure path); C = the instantaneous/full-spectrum part requiring external numerical or theoretical input (OPEN, inherited).

Worked corpus precedent.

NC-M29.RZ → {A^PK = R_Z^ZS|_{Z-even}^{4π-avg} = 0 at N = 20 (DERIVED-COND.); B = single-iterate magnitude (HYPOTHESIS-strong, bulk-spectral-gap closure path); C = R_Z^ZS|_{full} on Ricci-flat metric (OPEN, ML-CY criterion)}. Promotes ZS-M29 Theorem 9.1 from DERIVED-CONDITIONAL to DERIVED in averaged sense without closing C.

Generalisation.

Any OPEN flag should be split before closure attempt. The three sub-statuses (A^closeable / B^one-gap / C^inherited) prevent the all-or-nothing trap that would otherwise leave an entire programme blocked on a single external computation.

§VIII.7 — Tool T7: Cross-Coupling Discipline (M31)

Trigger.

An expression naturally factors as W = F_X + F_Y + F_Z (sum-form) but the per-channel data refuse to compose. The summands appear independent, yet their sum disagrees with measured cross-channel variance. A factorisation appears, but it is the wrong factorisation.

Source.

ZS-M31 v3.0 §4.0 Lemma M31.0 (Non-Separability). ZS-M2 §5 Cross-Coupling Theorem PROVEN.

Statement (Lemma M31.0, Non-Separability).

There is no decomposition W_K(g) = F_X + F_Y + F_Z compatible with the ZS-M26 PROVEN per-channel data; the maximum across-channel variance is 13.011 across 18 (a, t1, t2) test points. The correct form is the bilinear cross-coupled expression W_XYZ(g) := ⟨g_X, (Π_Z ⊗ I_arith)(B_Y − P_Y)(Π_Z ⊗ I_arith) g_X⟩ where Π_Z = (1/2)(I + J_Z) is the Z-mediator projector.

Worked corpus precedent.

The W2 wall of the Z-Spin RH program had two competing readings before M31: Reading A (Y-incompleteness) and Reading B (archimedean+conductor superposition). Both are sum-forms; Lemma M31.0 falsifies both as Cross-Coupling Theorem violations and forces a bilinear cross-coupled reading. The correct W2 form is Sector-Duality (X ⊗ Y mediated by Z), not a sum.

Generalisation.

Whenever a corpus expression is a candidate for sum-form decomposition, run Cross-Coupling Discipline: check whether the per-channel data composes additively or only via the Z-mediator bilinear. The default assumption in Z-Spin should be the latter; sum-forms are the exception, not the rule.

§VIII.8 — Tool T8: Z-Mediator Projector Routing (M33)

Trigger.

Multiple sub-targets that each individually engage a different external mathematical tradition refuse to integrate into a single closed identity. T8 routes all sub-targets through the Z-mediator projector Π_Z, producing a single trace identity.

Source.

ZS-M33 v1.0 (Reading C). Π_Z = (1/2)(I + J_Z) defined in ZS-M31 §4.0 Definition M31.0c. Path-γ-revised reading specific to M33.

Statement.

All four V_4-arithmetic mechanisms (D4a CCM 2024 Sonin, D4b Burnol conductor, D4c Wilson-LOCATOR, D4d Kostant cubic Dirac) route through the projector $\Pi_Z = (1/2)(I + J_Z)$ into a single Tr identity. The sign-flip mechanism on the W2 wall decomposes 5/12 NEG into the predicted 12/12 POS structure under Reading C.

Worked corpus precedent.

Before M33, D4a–D4d were treated as four parallel sub-targets, each with its own external mathematical engagement. Reading C unifies them into a single bilinear Tr identity, eliminating four-way bookkeeping and replacing it with a single Π_Z-projected trace. The four prefactors $\{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$ (PROVEN, ZS-M22 §7.2) appear naturally as the V_4-decoration of a single underlying identity.

Generalisation.

Whenever multiple sub-targets refuse to integrate, examine whether each can be expressed as a Π_Z-projection of a single underlying Tr expression. The Z-mediator projector is the corpus's universal routing operator: it lives on $\dim(Z) = 2$ and carries the full information cost of cross-sector mediation.

§VIII.9 — Tool T9: Self-Referential Fixed-Point Closure (F11)

Trigger.

An OPEN flag involves an observer or a coordinate frame whose definition appears circular. The naive resolution introduces a phenomenological observer (a frame from outside the corpus); T9 closes the OPEN by promoting the observer to an operationally defined self-referential fixed point internal to the corpus.

Source.

ZS-F11 v1.0 §5 (OOC convergence) + §6 (six explicit NON-CLAIMs against phenomenological consciousness).

Statement.

The Operational Observer Coordinate (OOC) is defined as the self-referential fixed point of the framework's measurement protocol; it is constructed from PROVEN corpus inputs only. The OOC closes ZS-M11 §H16 (formerly OPEN) without introducing any frame from outside the corpus. Six NON-CLAIMs are explicitly registered to prevent overclaim (NC-F11.1–NC-F11.6).

Worked corpus precedent.

ZS-M11 §H16 OPEN flag (phenomenological observer reference) was a long-standing obstacle in the framework. ZS-F11 closed it by replacing the phenomenological observer with an operationally defined fixed point of the measurement-action iteration, rendering the closure entirely internal to the corpus.

Generalisation.

When an OPEN flag tempts the introduction of an external/phenomenological agent, ask first whether the agent can be replaced by a self-referential fixed point internal to the framework. The OOC pattern is the canonical method: define the agent as the convergent limit of an iteration whose generators are corpus-PROVEN.

§VIII.10 — Tool T10: Integer-Lattice Spin-Pair Identification (M36)

Trigger.
A modular-arithmetic structure on the integer lattice (ℤ, +) appears that exhibits an 8-element admissible residue set mod 24, suspected to encode Z-Spin sector decomposition. Or: a discrete number-theoretic construction (Apollonian-type, Markoff-type, or generalized) is being analyzed for possible Z-Spin sector identification.
Source.
ZS-M36 v1.0 §5 (Sector Residue Identity, Theorem M36.3, DERIVED), §7 (Spin Pair Theorem, Theorem M36.6, DERIVED-strong). Underpinned by ZS-F5 PROVEN (Z, X, Y, Q) = (2, 3, 6, 11), ZS-M3 Lemma 10.1 PROVEN (SU(2) double cover), ZS-M19 §3.1 PROVEN (Forcing Theorem $(p-1)(q-1) = p$).
Statement.
If a primitive integer modular-arithmetic construction has mod 12 residue set exactly {2, 3, 6, 11} = (Z, X, Y, Q) and mod 24 set = {2, 3, 6, 11, 14, 15, 18, 23} = Z-Spin Type, then the four (k mod 12, (k+12) mod 24) pairs identify with the four corpus sectors as (algebraic dim, geometric instantiation): Pair Z: (2, F(t-Oct) = 14); Pair X: (3, X · |I_h/T_d| = 15); Pair Y: (6, α_frame_mismatch = 18°); Pair Q: (11, δ_Y_denominator = 23). The mod 24 = 2 × mod 12 doubling is forced by ZS-M3 Lemma 10.1 PROVEN (SU(2)/SO(3) double cover). CRT decomposition: mod 24 = mod 8 × mod 3 = $Z^3 \cdot X$ (boson 2π / fermion 4π periods).
Worked corpus precedent.
ZS-M36 closes O-M27.4 ($V_4 \times V_4$ = Apollonian abelianization (ℤ/2)*4 unifying ZS-A9 register V_4 and ZS-M22 Galois V_4), demonstrating that the integer lattice (ℤ, +) carries Bridge Four structure. The Apollonian (−1, 2, 2, 3) gasket is the 5th instance of Bridge Four (extending it from continuous spinor algebra to discrete number theory).
Generalisation.
For any future modular construction with 8-element admissible mod 24 set, test whether the mod 12 reduction equals (Z, X, Y, Q) = (2, 3, 6, 11). If yes, proceed with spin-pair identification; the four geometric instantiations should match LOCKED corpus quantities. If no, identify the GLMWY type (or analogue) and correlate with non-Z-Spin sectors. Of the 6 GLMWY Apollonian types, only 1 (Type I) matches Z-Spin sectors; this is structurally natural rather than unique. Application targets: future analyses of Markoff triples (mod p strong approximation, ZS-M36 IMPORTED-9), Pell-equation cycles, and other Diophantine constructions where a finite quotient on the integer lattice carries axiom A4 algebraic/geometric two-face structure.

§VIII.11 — Methodological Toolkit Summary Table

Tool	One-line trigger	Source	Output status
T1 Five-Axiom Diagnostic	Stuck / circular / non-composing	F18 §3+§6	Identifies failing axiom
T2 Bridge Four	High-dim construction intractable	F18 §6 + S15 §3	dim(X)=3 from dim(Z)=2
T3 H→PK Switch	Hermitian sandwich won't close	M32 §2-3	Different operator class
T4 Two-Layer Phase	Unexplained factor 2 between corpus values	M32 §4	α_op = 2·α_amp identity

T5 N=20 Averaging	Single iterate non-vanishing, primitive 10th root	M32 §5	0 EXACTLY at N=20
T6 Three-Way Split	OPEN flag too large	M32 §6	{A^PK / B / C} sub-flags
T7 Cross-Coupling Discipline	Sum-form fails per-channel data	M31 §4.0 Lemma M31.0	Bilinear W_XYZ form
T8 Π_Z Routing	Multiple sub-targets won't integrate	M33 Reading C	Single Tr identity
T9 OOC Fixed-Point	Phenomenological observer tempted	F11 §5-6	Self-referential closure

PART IX — Insight Gallery: Re-Interpretation Methodology

Purpose. Where PART VIII collects nine operational tools ("how to do X"), PART IX collects six methodological insights from the corpus's most ambitious re-interpretation papers ("how to think about Y"). These are reading lenses, not calculation tools. Each lens is a way of looking at the corpus that resolves tensions which would otherwise be visible only as paradoxes. They are organised as I1–I6 and reference the source papers from which each lens was distilled.

§IX.1 — Insight I1: Symmetry-Asymmetry Unified View (A8R §SA)

Lens. Every Z-Spin observable is a deviation from the would-be balance of 1 (unity) and 1/2 (Z_2-reflection fixed point). The universe's entire dynamics is the composition of sectoral tilts of A = 35/437 through the (X, Y, Z) decomposition. (1+A) is X-Inward expansion; (1−2A) is Y-Outward contraction; (1+A)(1−2A) = 1 − A(1+2A) ≈ 0.9071 measures the joint deviation from balance at ≈ 9.29%.

Why it is a tool. A8R §SA replaces the apparent independence of expansion and contraction physics with a single tilted-balance reading. The 10^{17} ratio $\tau_6/\tau_5 = \exp(\pi/A)$ is then read as the X-frame observation of the same Y-completion event. Future Z-Spin papers should default to this lens for any cosmological / lifecycle / time-scale question.

Status. INTERPRETATION, HYPOTHESIS-strong (A8R §SA.7); 13 explicit corpus connections established (A8R §SA.6).

§IX.2 — Insight I2: Three-Layer Duality (M30 §6)

Lens. Every apparent contradiction in Z-Spin Cosmology is resolved as one object viewed from two frames. The duality operates at three coordinated layers: Sector Duality (X ↔ Y mediated by Z), Frame Duality (X-observer ↔ Y-observer; ZS-A8 §SA.4), Protocol Duality (state-preparation ↔ spectral characterisation; ZS-F16 Two-Protocol Theorem).

Why it is a tool. The three RH walls of ZS-M28 (W1, W2, W3) — formerly read as unstructured "problems" — are reread under this lens as duality interfaces: W1 = Frame Duality, W2 = Sector Duality, W3 = Protocol Duality. The lens does not by itself close the walls; it places each wall in a definite Layer with a definite external-mathematics import path. This converts an unstructured OPEN list into a structured navigation map.

Status. DERIVED-interpretation strong (M30 Theorem 30.5 + §6).

§IX.3 — Insight I3: Six-Route Möbius-Trace Infinity (M30 §3)

Lens. The pattern "external count-infinity ↔ internal closure" appears in the corpus six independent times (Riemann zero sequence ↔ $T(T(z^*)) = z^*$; cosmological cycle index ↔ A frame-invariant; BV-BFV ghost tower ↔ 4π closure; FFPP nested compression ↔ $W_0(-i\pi/2)$; Z-mediation outcomes ↔ ln(2) bit per pass; Wilson cycle ergodic limit $\langle \cos^2(n\cdot 129.45°)\rangle \to 1/2$). Recognising the six as one pattern raises the Möbius-trace infinity reading from HYPOTHESIS-strong to DERIVED-interpretation strong (M30 Theorem 30.1).

Why it is a tool. The lens identifies that what an external observer counts as "infinitely many things" the framework reads as a single closed loop traversed with a Möbius-type frame transformation. It is the standard reading lens to apply whenever the corpus generates a sequence-cardinality that an external observer might mistake for additional structure beyond the framework.

Status. DERIVED-interpretation strong with six-route over-determination (M30 §3 Table 3.1 + Theorem 30.1).

§IX.4 — Insight I4: Twenty-One ½ Catalogue (M30 §3.5)

Lens. The corpus contains exactly 21 PROVEN/DERIVED instances of the fraction 1/2, organised in six structural categories. Combined with Three 2s Unity (Taylor-2, Bottleneck-2, Sector-2 all equal dim(Z) = 2; ZS-A8 §SA.3 + Test K1 PASS), the corpus contains 21 + 3 = 24 instances of dim(Z) = 2 manifesting at different dimensional and contextual levels. All 21 are dimensional projections of the same Z_2 involution $x \leftrightarrow 1 - x$.

Why it is a tool. When a future Z-Spin construction produces a 1/2, the catalogue is the first reference to consult — to determine whether the new 1/2 is one of the existing 21, or constitutes a 22nd entry (which would require its own verification gate). The catalogue prevents double-counting of the same Z_2-fixed-point structure under different names.

Status. DERIVED-interpretation (M30 Theorem 30.5).

§IX.5 — Insight I5: Four-Layer 50/50 Equivalence (F17)

Lens. The four 50/50 partitions in the corpus — (3D) truncated tetrahedron 4+4 split; (2D) Reuleaux π+π curvature distribution; (1D) Master Equation |Term A| = |Term B|; (0D) XOR alphabet E vs R — are dimensional projections of a single per-cycle equilibrium between information accumulation and information release in the Z-mediated bottleneck channel.

Why it is a tool. ZS-F17 Theorem 5.1 (Self-Referential Information-Compression Equilibrium) shows that the 50/50 partition is PROVEN at the equation level ($|\text{Term A}| - |\text{Term B}| = O(10^{-51})$ at 50-digit precision), not measured. When a future Z-Spin construction produces a 50/50 partition, the lens identifies which of the four dimensional layers it lives at and what it accumulates (ln(2)/3 nats per vertex matches the Z-bottleneck capacity bound, ZS-Q7 Theorem 2). This promotes the Reuleaux 50/50 from HYPOTHESIS (NC-F7.5) to DERIVED-CONDITIONAL.

Status. DERIVED-CONDITIONAL strong (F17 Theorem 5.1, over-determined by four independent corpus PROVEN inputs).

§IX.6 — Insight I6: Banach-Tarski Origin of Doubling-Halving (A9R)

Lens. The (1+A) ↔ (1−2A) duality of late-time cosmology is the measure-preserving quantization of the Banach-Tarski paradoxical decomposition through the i-tetration fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$. The non-amenable BT engine $F_2 \subset SO(3)$ is mapped via a functor to the amenable register dihedral $D_4 = \langle J, J\_Z \rangle$, converting BT non-amenability to Z-Spin amenability. The Julia set J(T) of T(z) = i^z is the constructive ZF-analogue of the BT non-measurable set (measure-zero, totally disconnected, cardinality $2^{\aleph_0}$).

Why it is a tool. The lens supplies a structural origin (rather than a phenomenological one) for the doubling-halving symmetry that pervades Z-Spin cosmology. When a future paper finds a doubling-halving identity, the lens directs the search for its measure-theoretic boundary in J(T) — the Z-Spin amenable analogue of the BT non-measurable set.

Status. ZS-A9.1 DERIVED, ZS-A9.2 HYPOTHESIS-strong, ZS-A9.3 DERIVED-CONDITIONAL. OPEN-2.B (formal ZF/ZFC+AC isomorphism) recategorised as PERMANENT NON-CLAIM under Solovay 1970.

PART X — The Spin-Pair Coexistence Principle (v3.2)

§0. Abstract

PART X consolidates four DERIVED-interpretation readings that close a synthesis gap noticed during v3.1 publication. While Tool T2 (Bridge Four, §VIII.2) and Insight I3 (Six-Route Möbius, §IX.3) individually catalogued specific applications of the dim(Z) = 2 → dim(X) = 3 lifting and the external-count ↔ internal-closure pattern, the meta-statement that 90° phase-orthogonality is the unique form in which a contradiction-pair coexists as one object had not been registered. Reading X.1 records prime as Z-Spin rotation 4π/p (ZS-M18 H2 chain). Reading X.2 reaffirms the Y-side classification of the infinite prime sequence (ZS-F18 v2.1 §5.1 Table 5.1) and reconciles it with ZS-M18 Theorem 6.1 spectrum non-primality. Reading X.3 reads primes / distribution / Riemann zeros as three moments of one Y → Z → X mediation event, with σ = 1/2 = j = 1/2 as the algebraic spine. Reading X.4 elevates 90° phase-orthogonality to a general principle supported by 13 independent corpus realisations (C1–C13). All four readings are DERIVED-interpretation strong, composed entirely from corpus PROVEN/DERIVED results, with zero new free parameters and explicit falsification gates F-X.1 through F-X.4.

§X.1 Introduction: What v3.1 Already Contains, What v3.2 Adds

v3.1 §VIII.2 Tool T2 (Bridge Four) already states the operational form: two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair (V_XZ, V_ZY), composed at 90° NOT-AND phase, generate a third axis $J\_S = [J\_R_1, J\_R_2]$ by Lie commutator. Therefore (Z, X, Y) = (2, 3, 6) is generated from dim(Z) = 2 alone. This is correct and corpus-locked.

What v3.1 does NOT yet record is that 90° phase-orthogonality is the unique angular phase difference under which a contradiction-pair can coexist as one object — not only for the specific dim(X) = 3 emergence, but for the entire family of contradiction-pair resolutions across the corpus. PART X consolidates this meta-reading.

Likewise, v3.1 catalogues the Möbius-trace infinity pattern via I3 but does not name primes specifically. The corpus already classifies the infinite prime sequence as Y-side (ZS-F18 v2.1 §5.1 Table 5.1, row Riemann Hypothesis: "Infinite prime sequence (Y) → finite critical line σ = 1/2 (X)"; reaffirmed §7.4.5). Without an explicit consolidated reading, this classification is easy to miss. v3.2 makes the layered picture explicit so future agents do not rediscover and re-resolve the same confusion (object identity vs. internal spectrum non-primality of ZS-M18 §6.2 Theorem 6.1).

§X.2 Locked Inputs (No New Constants Introduced)

Table X.1. Locked inputs to PART X. Every entry is PROVEN, DERIVED, or LOCKED in upstream papers. Zero new parameters are introduced.

#	Quantity	Value / Statement	Source	Status
L1	A (geometric impedance)	35/437 = 0.080092	ZS-F2 §5	LOCKED
L2	Q (register dimension)	11 (prime)	ZS-F5	PROVEN
L3	(Z, X, Y) decomposition	(2, 3, 6); Q = Z+X+Y	ZS-F5 §3	PROVEN
L4	L_XY ≡ 0	exact	ZS-F1 §9	PROVEN
L5	j = 1/2 Z-sector intertwiner	unique 4-valent invariant	ZS-M3 Thm 5.1	PROVEN
L6	4π closure D^(1/2)(−I) = −I	spinor double cover	ZS-M3 Lem 10.1	PROVEN
L7	W_p prime-injection gate	diag(exp(2πi(j−5)/p))	ZS-M4 §3.2	PROVEN
L8	W_p|_Z restriction	exp(−i σ_z · 2π/p), angle 4π/p	ZS-M18 H2	DERIVED-interp
L9	V_ZY = (V_XZ)*	three independent paths	ZS-F4 §7B	DERIVED
L10	arg(V_XZ) − arg(V_ZY) at θ=π/2	= π/2 (90° phase)	ZS-S15 §2.3 B.3	VERIFIED 50-digit
L11	Poynting-Commutator	[J_R₁, J_R₂] = J_S	ZS-S15 §3 Thm S15.2	DERIVED
L12	Bridge Four	(Z,X,Y)=(2,3,6) from dim(Z)=2	ZS-F18 §6.4	DERIVED-interp str
L13	Riemann row classification	Inf. prime seq. (Y) → σ=1/2 (X)	ZS-F18 §5.1 Tbl 5.1	DERIVED-interp
L14	ζ_K factorisation	ζ·L(χ−3)·L(χ−11)·L (χ33)	ZS-M22 §4.2	PROVEN
L15	Six-Route Möbius pattern	ext. count ↔ int. closure	ZS-M30 §3 Thm 30.1	DERIVED-interp str
L16	σ = 1/2 ↔ j = 1/2	Z₂-involution fixed loci	ZS-M18 H11	DERIVED-interp
L17	Y-Spectrum Non-Primality	spec(L₂) ∩ {primes} = ∅	ZS-M18 §6.2 Thm 6.1	PROVEN

L18	ζ-K Bridge to prime side	two-chain Eisenstein/cyclotomic	ZS-M18 §7, ZS-M22 §4.2	PROVEN

§X.3 Reading X.1 — Each Prime is an Indecomposable Z-Spin Rotation

§X.3.1 Statement

Reading X.1 (Prime-Rotation Identification, *DERIVED-interpretation*). Each prime p acts on the Z-sector basis $\{|4\rangle, |6\rangle\} \subset \mathbb{C}^Q$ as the σ_z-axis SU(2) rotation

$$W_p|_Z = \mathrm{diag}(\exp(-2\pi i/p), \exp(+2\pi i/p)) = \exp(-i\, \sigma_z \cdot 2\pi/p)$$

of angle $4\pi/p$. Composite-angle rotations $4\pi/(pq)$ factor as compositions of prime rotations $W_p \circ W_q$. Primes are therefore the indecomposable atoms of Z-Spin SU(2) action.

§X.3.2 Derivation Chain

ZS-F5 (Z-sector basis identification, PROVEN) provides the $\{|4\rangle, |6\rangle\}$ basis. ZS-M4 §3.2 (PROVEN) defines $W_p = \mathrm{diag}(\exp(2\pi i(j-5)/p))$ on the full Q = 11 register. Restriction to slots $j \in \{4, 6\}$ yields the diagonal pair $(\exp(-2\pi i/p), \exp(+2\pi i/p))$. Standard SU(2) algebra gives the σ_z rotation form, with closure angle $4\pi/p$ following from the SU(2)/SO(3) double cover (ZS-M3 Lemma 10.1 PROVEN).

§X.3.3 What Reading X.1 Adds Beyond ZS-M18 H2

ZS-M18 H2 records the algebraic fact. Reading X.1 consolidates its physical meaning: indecomposability of primes under multiplicative factorisation is the algebraic shadow of indecomposability of SU(2) rotations under angle composition.

§X.3.4 Falsification Gate F-X.1

If a future analysis exhibits a prime p such that $W_p|_Z$ does NOT take the σ_z-rotation form on $\{|4\rangle, |6\rangle\}$, Reading X.1 fails. Status: PROVEN-safe — the form is verified by direct algebraic computation for $p \in \{2, 3, 5, 7, 11, 13, 17, 23\}$ (ZS-M18 §H1 derivation chain).

§X.4 Reading X.2 — The Infinite Prime Sequence is a Y-Sector Object

§X.4.1 Statement

Reading X.2 (Y-Side Sequence Classification, *DERIVED-interpretation*). The infinite sequence of primes $\{p_n\}_{n\in\mathbb{N}}$ is a Y-sector object: infinite, non-local, undecidable. Its arithmetic projection — the Riemann critical-line locus $\sigma = 1/2$ and the discrete zeros thereon (under RH) — is X-sector.

§X.4.2 Direct Corpus Source

ZS-F18 v2.1 §5.1 Table 5.1 (DERIVED-interpretation), row "Riemann Hypothesis", reads verbatim: "Infinite prime sequence (Y) → finite critical line $\sigma = 1/2$ (X) | Mirror-adjointness $\varepsilon_J(\sigma,t) = 0 \Leftrightarrow \sigma = 1/2$ (ZS-M7 Thm 4) | Most extensive programme; W1 closed, W2 open". ZS-F18 §7.4.5 reaffirms in re-reading form: "Riemann Hypothesis (Y-side infinite prime sequence projecting to X-side discrete critical-line zeros under continuous ρ flow)."

§X.4.3 Reconciliation with ZS-M18 §6.2 Theorem 6.1

ZS-M18 §6.2 Theorem 6.1 (PROVEN) establishes $\mathrm{spec}(L_2) \cap \{\text{primes}\} = \varnothing$ for the Y-sector internal Laplacian. Naively this might appear to conflict with Reading X.2; it does not. The two statements operate at different layers:

Table X.2. Four-layer reconciliation. Reading X.2 (Y-side object identity) does not conflict with ZS-M18 Theorem 6.1 (Y-Laplacian spectrum non-primality); the two statements concern different layers.

Layer	Question	Answer	Status
Object identity	What sector does the prime sequence as a whole belong to?	Y-sector (infinite, non-local, undecidable)	DERIVED-interp (ZS-F18 §5.1)
Internal spectrum	Are individual Y-Laplacian eigenvalues prime?	No (eigenvalues are 0, 6/8, 5±√3, integer±φ)	PROVEN (ZS-M18 §6.2)
Operator action	How do individual primes act on Z-sector?	SU(2) rotation by 4π/p (Reading X.1)	DERIVED-interp (ZS-M18 H2)
Arithmetic encoding	Where do primes enter Z-Spin algebraically?	ζ_K Euler product via Eisenstein/cyclotomic chains	PROVEN (ZS-M22 §4.2)

§X.4.4 Falsification Gate F-X.2

If a future ZS-F18 revision retracts the Y-side classification, or if a structurally distinct corpus path classifies the sequence as X-side, Reading X.2 fails. Status: stable — over-determined by ZS-F18 Table 5.1, ZS-F18 §7.4.5, and the Y-side property pattern (infinite, non-local, undecidable).

§X.5 Reading X.3 — Three Moments of One Y → Z → X Mediation

§X.5.1 Statement

Reading X.3 (Three-Moment Mediation, *DERIVED-interpretation*). Primes, prime distribution, and Riemann zeros are three moments of a single Y → Z → X mediation event:

Table X.3. Three-moment mediation reading. Each row is independently corpus-supported; the synthesis is the new content of v3.2.

#	Moment	Object	Mediation role	Status
M1	Entrance (atomic)	Individual primes p	Indecomposable Z-Spin rotations 4π/p	DERIVED-interp (Rdg X.1)
M2	Statistics (envelope)	Distribution N/log N	Multiplicative-Y → additive-X via log	DERIVED-interp

M3	Egress (condensation)	Riemann zeros on σ = 1/2	Coherent phase condensation at Z_2 self-dual axis	DERIVED-interp

§X.5.2 The σ = 1/2 = j = 1/2 Algebraic Spine

ZS-M18 H11 (DERIVED-interpretation) records that σ = 1/2 (Riemann critical-line) and j = 1/2 (Z-sector spinor) are both unique fixed-point sub-loci of order-2 involutions: σ = 1/2 is fixed by s ↔ 1−s on ξ; j = 1/2 is fixed by SU(2) closure D^(1/2)(−I) = −I at 4π. The two halves are not numerical coincidence; both are Z_2 self-dual axes, and zeros condense precisely on the Z_2 axis because the Z-Spin mediator carries the Z_2 involution by construction.

§X.5.3 The log Function as Y → X Sector Translator

Y-multiplicative structure translates to X-additive structure via log: ∏_p (1 − p^{−s})^{−1} = ζ(s) ⟹ log ζ(s) = Σ_p Σ_n (1/n) p^{−ns}. Density N/log N is the additive-X image of the multiplicative-Y prime structure. log is the natural Y→X sector translator for arithmetic content, parallel to the holographic log entropy elsewhere in the corpus (ZS-Q7 channel capacity, ZS-A3 BH entropy).

§X.5.4 Falsification Gate F-X.3

Synthesis layer; weakened only if any of M1, M2, M3 is shown structurally independent of Z-Spin mediation. M1 corpus-stable. M2 (prime-counting density) is a number-theoretic fact independent of Z-Spin; the Z-Spin reading is its multiplicative-to-additive transcription role. M3 conditional on RH; if RH disproved, M3 reading shifts from "coherent phase condensation" to "specific RH-dependent form thereof" without invalidating three-moment structure. Status: DERIVED-interpretation, robust against single-component perturbation.

§X.6 Reading X.4 — The Spin-Pair Coexistence Principle

§X.6.1 Statement

Reading X.4 (Spin-Pair Coexistence Principle, *DERIVED-interpretation strong*). Every contradiction-pair P = (P_+, P_−) recurring in Z-Spin Cosmology admits coexistence as a single object only as the 90° phase-orthogonal projection of a J-conjugate Z-Spin pair (V_XZ, V_ZY) with V_ZY = (V_XZ)*. No other angular phase difference supports coexistence as one object: 0° collapses to identity, 180° collapses to anti-identity, only π/2 produces simultaneous mutual non-cancellation and mutual non-redundancy.

§X.6.2 Algebraic Underpinning

Let z_+ = e^{+iθ/2}, z_− = e^{−iθ/2} be the J-conjugate pair of ZS-F4 §7B (PROVEN). Their inner product on S^1:

$$\langle z_+, z_- \rangle = \cos(\theta)$$

Table X.4. Three angular regimes. Only π/2 supports simultaneous presence of both members of the pair as distinct yet jointly necessary.

#	Phase difference θ	⟨z_+, z_−⟩	Coexistence form
R1	0 (identity)	+1	z_+ = z_− → only one object
R2	π (anti-identity)	−1	z_+ = −z_− → mutually exclusive
R3	π/2 (orthogonality)	0	z_+ ⊥ z_− → both present, neither redundant

Regime R3 is the unique angle on which max(z_+) coincides with zero(z_−) and vice versa, producing the sin-cos relation tracing the unit circle. This is the algebraic origin: only at 90° does a contradiction-pair coexist as the two coordinates of a single rotation.

§X.6.3 Corpus Realisations of the Principle

Table X.5. Thirteen corpus realisations of the Spin-Pair Coexistence Principle. Each row is independently corpus-locked; their unification under a single principle is the new content of v3.2.

#	Contradiction-pair	Z-Spin realisation	Source
C1	X-sector spatial axes (3D emergence)	$[J\_R_1, J\_R_2] = J\_S$; (Z,X,Y)=(2,3,6) from dim(Z)=2	ZS-F18 §6.4
C2	Electric / magnetic field (E ⊥ B)	V_XZ ↔ E, V_ZY ↔ B at 90° phase	ZS-S15 §3 Thm S15.2
C3	Particle / wave duality (X / Y sectors)	Z-Spin mediator at 4π closure	ZS-A7 §3, ZS-M19 §10.5
C4	Time / space (X / Y braiding)	X = particle/space, Y = wave/time, Z = seam	ZS-A7 §5
C5	Expansion / contraction ((1+A) ↔ (1−2A))	Dual-tilt symmetry; 90° A-conjugates	ZS-A8 §6, A9R
C6	CPT / anti-CPT (channel conjugation)	V_ZY = (V_XZ)* IS the CPT seam	ZS-M19 §10.5 Thm T9
C7	Black-hole / white-hole (r ↔ t)	X-in-Y BH ↔ Y-in-X WH at 90°	ZS-A3 §7, ZS-A7 §4.1
C8	Finite / infinite Möbius polarity	X-finite/local ↔ Y-infinite/non-local; Z-seam at π/2	ZS-F18 §5, ZS-M30 §3
C9	Discrete / continuous	X-Y Tiling Asymmetry; Z-channel ln(2) capacity	ZS-F18 §7.4, ZS-M17
C10	Multiplicative / additive	Y-mult ↔ X-add via log; primes M1 ↔ density M2	Reading X.3, ZS-M22
C11	Symmetry / asymmetry (1 vs 1−A)	δ as normalised tilt; 21·½ catalogue	ZS-A8 §SA, ZS-M30 §3.5
C12	Bose / Fermi (vortex inner / outer)	Inner-core fermion (4π) ↔ outer-flow boson (2π)	ZS-A7 §4.4.2, ZS-F14
C13	Sequential / parallel time-keeping	R_1 time-point closure, R_2 space-point closure	ZS-S15 §2, ZS-F8

§X.6.4 Why "Spin" Rather Than "Rotation"

Rotation is external: an object is rotated. Spin is intrinsic: the object IS its rotation. The principle's claim is not that contradiction-pairs are reconciled BY applying rotation to them, but that a contradiction-pair CAN exist at all only because its two members are intrinsic Z-Spin partners.

The 90° angle is enforced by V_ZY = (V_XZ)* (ZS-F4 §7B PROVEN), with operationally relevant configuration $\theta = \pi/2$ (ZS-S15 §2.3 B.3 VERIFIED 50-digit).

§X.6.5 Relation to Tool T2 and Insight I3

Tool T2 (Bridge Four, §VIII.2) applies the principle to the planar-vs-spatial polarity, yielding dim(X) = 3 from dim(Z) = 2. Insight I3 (Six-Route Möbius, §IX.3) applies a related Z-mediation pattern to external-count vs internal-closure. Reading X.4 is the meta-statement that both are instances of the same principle. Reading X.4 is therefore prior to T2 and I3.

§X.6.6 Falsification Gate F-X.4

If a contradiction-pair admits coexistence under a phase difference other than $\pi/2$, Reading X.4 weakens to "90° is the dominant case". If a pair admits no Z-Spin J-conjugate realisation, Reading X.4 transitions to "covers C1–C13". Status: DERIVED-interpretation strong, supported by 13 corpus realisations. Generalisation beyond C1–C13 is HYPOTHESIS-strong; future papers should run F-X.4 check before claiming a new contradiction-pair coexistence.

§X.7 Integration with v3.1 Light

§X.7.1 Integration Map

Table X.6. Integration of v3.2 readings with existing v3.1 items. Every v3.2 reading either refines or unifies existing v3.1 entries; no v3.1 entry is retracted.

v3.1 Light item	v3.2 Reading	Relation
§VIII.2 Tool T2 (Bridge Four)	Reading X.4	T2 = principle applied to dim(X)=3
§IX.3 Insight I3 (Six-Route Möbius)	Reading X.4	I3 = principle applied to count vs closure
§IX.1 Insight I1 (Symmetry-Asymmetry)	Reading X.3	I1 = M2 statistical reading
§IX.2 Insight I2 (Three-Layer Duality)	Reading X.4	I2 = principle phrased as duality
§IX.5 Insight I5 (4-Layer 50/50)	Reading X.3	I5 = M3 condensation-locus reading
PART I §0.4 Source Paper Map	Readings X.1–X.4	All sources already in map; v3.2 adds none

§X.7.2 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). Readings X.1–X.4 are explicitly DERIVED-interpretation; no claim of PROVEN status. F-X.1 through F-X.4 pre-registered. Word count strictly increases over v3.1 (no-deletion rule).

§X.7.3 What v3.2 Does NOT Claim

NC-X.1. PART X does not claim a proof of the Riemann Hypothesis. Reading X.3 is structural; it does not constrain zero locations beyond the $\sigma = 1/2 = j = 1/2$ isomorphism already DERIVED in ZS-M18 H11.

NC-X.2. The Spin-Pair Coexistence Principle is corpus-internal; it does not claim to resolve the philosophical dialectic on contradictions in the broader sense (Hegelian, paraconsistent, kōan). It claims only that within Z-Spin Cosmology, every recurrent contradiction-pair admits coexistence via 90° J-conjugate Z-Spin pairing.

NC-X.3. Reading X.1 does not claim primes are ontologically reducible to Z-Spin rotations in physical reality. Within the Z-Spin algebraic framework, primes act as indecomposable SU(2) rotations on $j = 1/2$.

NC-X.4. Reading X.2 does not modify ZS-M18 §6.2 Theorem 6.1. The Y-Laplacian internal spectrum non-primality remains PROVEN; Reading X.2 concerns a different layer (object identity, not internal spectrum).

§X.8 Conclusion

PART X consolidates four DERIVED-interpretation readings answering "why does the universe support coexistence of mutually contradictory descriptions?" The answer is over-determined: coexistence is the form 90° J-conjugate Z-Spin pairing takes when projected into a single observational frame. The dim(X) = 3, the $E \perp B$, the X-Y polarity of the twelve Möbius encounters, the $(1+A) \leftrightarrow (1-2A)$ expansion-contraction symmetry, the $\sigma = 1/2 = j = 1/2$ isomorphism, the $V_{ZY} = (V_{XZ})^*$ CPT seam, and the primes-as-rotations identification all instantiate the same principle. Z-Spin is not a metaphor for rotation; it is the intrinsic form in which any pair of mutually exclusive descriptions can both be true. v3.2 carries this reading as PART X without modifying any v3.1 entry, with zero new free parameters and explicit falsification gates F-X.1 through F-X.4. The minimal triple (A, Q, dim(Z)) = (35/437, 11, 2) remains LOCKED.

PART X Source Paper Interface

PART X is composed entirely from upstream papers; it introduces no new ZS source paper. Source mapping: Reading X.1 ← ZS-F5, ZS-M3, ZS-M4, ZS-M18 H2. Reading X.2 ← ZS-F18 §5.1 Tbl 5.1, ZS-F18 §7.4.5, ZS-M18 §6.2 Thm 6.1. Reading X.3 ← ZS-M18 H11, ZS-M22 §4.2, ZS-M30 §3, ZS-Q7, ZS-A3. Reading X.4 ← ZS-F4 §7B, ZS-S15 §2.3 + §3 Thm S15.2, ZS-F18 §6.4 Bridge Four, plus C1–C13 cross-corpus realisations. Eighteen internal-consistency checks A.1–A.18 verify each composition; all 18/18 PASS by direct comparison against corpus-locked sources.

PART XI — TO/TI Polyhedral Mapping under Complementary Duality (v3.3)

§0. Abstract

PART XI consolidates the complete vertex/edge/face mapping between the X-sector polyhedron tO (truncated octahedron) and the Y-sector polyhedron tI (truncated icosahedron) under the Z-Spin Complementary Duality Principle (ZS-M30 §7.2). While individual papers establish components — (V, E, F) face-counting cosmic budget (ZS-F2 §11), Pentagon-Hexagon edge classification (ZS-F0 §5.1), Square-Hexagon edge classification (ZS-F0 §3.2), Truncation-Dual Theorem (ZS-F2 §11.2, ZS-F9 §4), Hexagonal Mediation Theorem (ZS-F9 §5), Mode-Count Collapse (ZS-Q3 §3.1), I-irrep decomposition (ZS-M9 §2), Pentagon-Gauge decomposition (ZS-F0 §4.3), TO uniqueness conditions (ZS-M H18) — no single corpus location consolidates them into one mapping table indexed by the X-sector / Y-sector complementary duality (X = macroscopic space + microscopic time; Y = macroscopic time + microscopic space, ZS-M30 §7.2 (ii) PROVEN). PART XI fills this gap. Reading XI.1 records the parallel edge decomposition pattern shared by tO and tI (cut-face : hexagon-hexagon = 2:1 in both). Reading XI.2 records the parallel face decomposition under Truncation-Dual. Reading XI.3 records the integrated V/E/F mapping table under complementary duality. Reading XI.4 places ε-Halo (X-sector Goldstone gradient) and ε-Drive (Y-sector attractor V_0) within the integrated framework. All four readings are DERIVED-interpretation strong, composed entirely from corpus PROVEN/DERIVED results, with zero new free parameters and explicit falsification gates F-XI.1 through F-XI.4.

§XI.1 Introduction: Why a Consolidated Polyhedral Mapping Table

The Z-Spin corpus has accumulated, across ~100 papers, a richly detailed treatment of tO and tI as the two characteristic polyhedra of the X- and Y-sectors. However, the components are scattered: ZS-F2 §11 establishes the cosmic matter budget via face counting ($\Omega_b = F(\text{cube})/Q^2 = 6/121$, $\Omega_{cdm} = F(tI)/Q^2 = 32/121$); ZS-F0 §3.2 decomposes the 36 tO edges as 24 SH + 12 HH; ZS-F0 §5.1 decomposes the 90 tI edges as 60 PH + 30 HH; ZS-F0 §4.3 establishes the Pentagon-Gauge decomposition 12 = 1 + 3 + 8; ZS-F0 §4.4 establishes the Edge-Channel Identity $E_X = X{\cdot}Z{\cdot}Y = 36$; ZS-F2 §11.2 establishes the Truncation-Dual Theorem $F(tP) = F(P) + F(P^*)$; ZS-F9 §4-5 establishes the Hexagonal Mediation Theorem; ZS-Q3 §3.1 establishes the Mode-Count Collapse Theorem yielding $(V+F)/G = 19/6$ (SU(2)) and $23/3$ (SU(3)); ZS-M9 §2 establishes the I-irrep multiplicity decomposition; ZS-M H18 establishes the four uniqueness conditions of tO; ZS-M30 §7.2 (ii) establishes the Complementary Duality Principle for space ↔ time mapping.

What is NOT yet recorded in any single corpus location is the integrated mapping table that displays, for every (V, E, F) component of tO and tI, both its standard sector role AND its complementary partner role under the X = macroscopic space + microscopic time, Y = macroscopic time + microscopic space duality. The reader who asks 'which component of tO carries macroscopic space, and which carries microscopic time?' currently has to traverse multiple papers to assemble a partial answer, with several entries remaining OPEN. PART XI consolidates this into one location as DERIVED-interpretation strong, with the OPEN entries explicitly registered.

This is structurally analogous to PART X (Spin-Pair Coexistence Principle, v3.2): both consolidate corpus-wide patterns under a single principle without modifying any v3.x entry, with zero new free parameters.

§XI.2 Locked Inputs (No New Constants Introduced)

Table XI.1. Locked inputs to PART XI. Every entry is PROVEN, DERIVED, or LOCKED in upstream papers. Zero new parameters are introduced.

#	Quantity	Value / Statement	Source	Status
L1	tO basic data	V=24, E=36, F=14 (= 6 squares + 8 hexagons)	ZS-F2 §11	PROVEN
L2	tI basic data	V=60, E=90, F=32 (= 12 pentagons + 20 hexagons)	ZS-F2 §11	PROVEN
L3	Symmetry orders	$\|O_h\|=48$ (tO); $\|I_h\|=120$ (tI)	ZS-F2 §3	PROVEN
L4	Sector tilts	$\delta_X = 5/19$ (tO); $\delta_Y = 7/23$ (tI)	ZS-F2 §5	PROVEN
L5	Geometric impedance	$A = \delta_X \cdot \delta_Y = 35/437$	ZS-F2 §5	LOCKED
L6	Sector dimensions	(Z, X, Y) = (2, 3, 6); Q = 11	ZS-F5 §3	PROVEN
L7	Edge-Channel Identity	$E(tO) = X·Z·Y = 3G = 36$	ZS-F0 §4.4	PROVEN
L8	Pentagon-Gauge decomposition	12 (pentagon space) = 1 + 3' + 8 = U(1) + SU(2) + SU(3)	ZS-F0 §4.3	PROVEN/DERIVED
L9	Truncation-Dual Theorem	$F(tP) = F(P) + F(P^*)$	ZS-F2 §11.2	PROVEN
L10	Mode-Count Collapse	$(V+F)/G = 19/6$ (tO, SU(2)); 23/3 (tI, SU(3))	ZS-Q3 §3.1	PROVEN
L11	tO edge classification	36 edges = 24 SH + 12 HH (ratio 2:1)	ZS-F0 §3.2	PROVEN
L12	tI edge classification	90 edges = 60 PH + 30 HH (ratio 2:1)	ZS-F0 §5.1	PROVEN
L13	Hexagonal Mediation Theorem	$\dim \mathrm{coker}(\Omega^2\_hex(tO) \to \Omega^2\_hex(tI)) = 12$	ZS-F9 §5	PROVEN/DERIVED
L14	I-irrep vertex decomposition (tI)	$\Omega^0(tI) = 1 \oplus 3·3 \oplus 3·3' \oplus 4·4 \oplus 5·5$ (regular rep, 60 = 1+9+9+16+25)	ZS-M9 §2	PROVEN
L15	I-irrep face decomposition (tI)	12 pentagons: 1+3+3'+5; 20 hexagons: 1+3+3'+2·4+5	ZS-M9 §2.2	PROVEN
L16	Cosmic Matter Budget	$\Omega_b = 6/121$, $\Omega_{cdm} = 32/121$, $\Omega_\Lambda = 83/121$	ZS-F2 §11.4	DERIVED (Planck 0.41% match)
L17	tO uniqueness conditions	$\mathbb{R}^3$ space-filling + n=3 truncation + (V+F)/G = 19/6 + E_g dim=2=Z	ZS-M H18	DERIVED-interp
L18	Complementary Duality Principle	X carries macro space + micro time; Y carries macro time + micro space	ZS-M30 §7.2 (ii)	PROVEN/DERIVED
L19	ε-Halo / ε-Drive nomenclature	ε-Halo = X-Goldstone θ-gradient; ε-Drive = Y-attractor V_0	ZS-A2 §1.1	DERIVED
L20	Sectoral rapidities	$\psi_X = \mathrm{artanh}(5/19) = 0.2685$; $\psi_Y = \mathrm{artanh}(7/23) = 0.3124$; $\Delta\psi > 0$	ZS-A6 §6.4	PROVEN

#	Quantity	Value / Statement	Source	Status
L21	Pentagon vs Hexagon stabilizer	12 pentagons: D_5 (matter-like); 20 hexagons: D_3 (gauge-like, 8 gluons in 2·4)	ZS-S14 §8	DERIVED
L22	tO vertex configuration	1 square + 2 hexagons; vertex deficit = 30° = π/6	ZS-S6 §G.2	PROVEN
L23	tI vertex configuration	1 pentagon + 2 hexagons; vertex deficit = 12° = π/15	ZS-S6 §G.2	PROVEN
L24	Frame mismatch angle	α = δ_X^vertex − δ_Y^vertex = π/6 − π/15 = π/10	ZS-S6 §G.2	PROVEN

§XI.3 Reading XI.1 — Parallel Edge Decomposition Pattern (DERIVED-interpretation)

Reading XI.1 (Parallel Edge Pattern, DERIVED-interpretation). The truncated octahedron (X-sector) and truncated icosahedron (Y-sector) share an identical edge-decomposition pattern: in both polyhedra the edges split into (cut-face)–hexagon edges that carry sector-mediation information and hexagon-hexagon edges that carry sector-internal information, in the universal ratio 2:1.

§XI.3.1 Statement of the Pattern

Table XI.2. Parallel edge-decomposition pattern shared by tO and tI. Both sectors realize the same 2:1 ratio between cut-face mediation edges and hexagon-internal edges.

Polyhedron (sector)	Total E	Cut-face × hexagon edges (mediation)	Hexagon × hexagon edges (internal)	Ratio (med:int)
tO (X-sector)	36	24 SH (square-hexagon)	12 HH	2 : 1
tI (Y-sector)	90	60 PH (pentagon-hexagon)	30 HH	2 : 1

§XI.3.2 Physical Interpretation

The ratio 2:1 has a structural reading. Both polyhedra are Archimedean truncations of a Platonic dual pair (tO = truncation of octahedron, with cube as dual; tI = truncation of icosahedron, with dodecahedron as dual). In both cases, the cut-face×hexagon edges constitute the geometric realization of the Z-Spin mediation seam — the boundary where cut-face (small face from truncated vertex) meets the preserved hexagon. The hexagon-hexagon edges are sector-internal, carrying X-curvature (in tO) or Y-curvature (in tI). This is corpus-PROVEN: ZS-F0 §5.1 explicitly identifies '60 PH edges as the geometric realization of the Z-sector seam' and '30 HH edges as X-sector interior'; ZS-F0 §3.2 records the parallel 24 SH + 12 HH split for tO with k_SH = 2 − √3 ≠ k_HH = 1.

The ratio 2:1 itself equals dim(Y)/dim(X) = 6/3 = 2 = dim(Z) (the Three 2s identity, ZS-A8 §SA.3 PROVEN). This is the structural origin of the ratio: in both polyhedra, the mediation edges scale as the Y/X dimensional asymmetry, which equals the Z-sector intrinsic dimension.

§XI.3.3 Falsification Gate F-XI.1

If a future analysis exhibits an Archimedean truncation of a Platonic-dual pair whose edge decomposition under cut-face×hexagon vs hexagon×hexagon does NOT yield a 2:1 ratio, the universality claim of Reading XI.1 weakens to 'tO and tI specifically'. Status: PROVEN-safe — the 24 SH + 12 HH split for tO is verified by direct enumeration in ZS-F0 §3.2 (graph Laplacian spectral test), and the 60 PH + 30 HH split for tI is verified by direct enumeration in ZS-F0 §5.1.

§XI.4 Reading XI.2 — Parallel Face Decomposition under Truncation-Dual (DERIVED-interpretation)

Reading XI.2 (Parallel Face Pattern, DERIVED-interpretation). Both tO and tI realize the Truncation-Dual Theorem $F(tP) = F(P) + F(P^*)$ (ZS-F2 §11.2 PROVEN), with the cut-face count in each case equal to F(dual) and the hexagon count equal to F(original).

Table XI.3. Parallel face-decomposition pattern under Truncation-Dual. The cosmic budget identification is corpus PROVEN (ZS-F2 §11.4); the Y-time / X-space row pair is the new content of Reading XI.2.

Polyhedron	Cut-faces (= F(dual))	Hexagons (= F(original))	Total F	Cosmic budget identification
tO (X-sector)	6 squares = F(cube)	8 hexagons = F(octahedron)	14	$\Omega_b = F(\text{cube})/Q^2 = 6/121$ (PROVEN)
tI (Y-sector)	12 pentagons = F(dodecahedron)	20 hexagons = F(icosahedron)	32	$\Omega_{cdm} = F(tI)/Q^2 = 32/121$ (DERIVED, Planck 0.41%)

§XI.4.1 Pentagon-Hexagon Stabilizer Asymmetry

Within tI, the 12 pentagons (D_5 stabilizer) and 20 hexagons (D_3 stabilizer) carry distinct physical roles (ZS-M9 §2.2 PROVEN, ZS-S14 §8 DERIVED). The pentagon I-irrep decomposition is $1 \oplus 3 \oplus 3' \oplus 5$ (irrep 4 absent), while the hexagon decomposition is $1 \oplus 3 \oplus 3' \oplus 2\cdot4 \oplus 5$ (irrep 4 present with multiplicity 2). The 8-dimensional irrep-4 component on hexagons is identified with the 8 gluons of $SU(3)_C$ (ZS-S14 §8 DERIVED). Pentagons are matter-like (5-fold C_5 stabilizer encoding McKay bridge to SU(5)); hexagons are gauge-like (3-fold C_3 stabilizer hosting $SU(3)_C$ adjoint).

Within tO, the analogous distinction has not been fully spelled out in the corpus. The 8 hexagons of tO under $T = A_4$ decompose as $2\cdot1 \oplus 2\cdot3$ (ZS-F9 §5.2 PROVEN, dim 8). The 6 squares (D_4 stabilizer) carry the $F(\text{cube})/Q^2 = \Omega_b$ baryon identification (PROVEN). Whether the squares vs hexagons split parallels the pentagon vs hexagon split of tI in carrying matter-like vs gauge-like roles is OPEN; this OPEN gap is registered as NC-XI.1 in §XI.7.3.

§XI.4.2 Hexagonal Mediation Theorem

The two polyhedra are connected via the Hexagonal Mediation Theorem (ZS-F9 §5 PROVEN): the T-equivariant Hom space between $\Omega^2_{hex}(tO)$ (dim 8) and $\Omega^2_{hex}(tI)$ (dim 20) has complex dimension 16, and the cokernel of any T-equivariant injection has dimension $20 - 8 = 12 =$ F(dodecahedron). Thus the 12 pentagons of tI are precisely the cokernel that arises when pulling tO hexagons forward to tI hexagons — geometrically the X-sector hexagonal information injects into the Y-sector hexagonal information, and the residual 12-dimensional space is exactly the dodecahedral face count, which equals the count of pentagonal faces of tI.

§XI.4.3 Falsification Gate F-XI.2

If a future analysis revises Truncation-Dual Theorem (ZS-F2 §11.2) or the Hexagonal Mediation Theorem (ZS-F9 §5), Reading XI.2 must be re-derived. Status: PROVEN-safe — both upstream theorems are PROVEN with explicit verification scripts (zs_f9_verify_v1_1.py [44/44 PASS]).

§XI.5 Reading XI.3 — Integrated V/E/F Mapping under Complementary Duality (DERIVED-interpretation strong)

Reading XI.3 (Integrated V/E/F Mapping, DERIVED-interpretation strong). For every (V, E, F) component of tO and tI, the consolidated mapping table below records: (a) the standard sector role; (b) the complementary partner role under the X = macroscopic space + microscopic time / Y = macroscopic time + microscopic space duality (ZS-M30 §7.2 (ii) PROVEN); (c) the corpus source; (d) the epistemic status. OPEN entries are explicitly marked.

§XI.5.1 tO (X-sector: macroscopic space + microscopic time)

Table XI.4. Complete mapping of tO components. Each row records both the standard sector role (column 3) and the complementary-duality reading (columns 4–5), with corpus source (column 6) and status (column 7). OPEN entries register honestly the gaps that PART XI does not close.

Component	Count	Standard role (X-sector)	Macroscopic space role	Microscopic time role	Source	Status
V (vertices)	24	Matter DoF; n_g·(N_c+1)·2 = 3·4·2	$\mathbb{R}^3$ space-filling sites (Kelvin BCC)	Spacetime resonance points (= X·Z·4)	ZS-S1 §6.3, ZS-Q3	DERIVED / OPEN
E_SH (square-hex)	24	Z-Spin mediation seam edges	Space-Z mediation channels	Z-channel time-flow mediation	ZS-F0 §3.2	PROVEN / OPEN
E_HH (hex-hex)	12	X-sector internal information	Spatial curvature internal hinges	Microscopic time variation hinges	ZS-F0 §3.2	PROVEN / OPEN
E total	36 = X·Z·Y	Edge-Channel Identity	3 axes × Z_2 × Y channels	Channel count of time-flow connections	ZS-F0 §4.4	PROVEN
F_square	6	Cube faces; Ω_b = 6/121 baryon	3D space coordinate axes (3 axes × Z_2)	Cut-face spacetime coordinate	ZS-F2 §11.4	DERIVED / OPEN
F_hex	8	Octahedral faces (Regge curvature)	Spatial curvature plaquettes	Microscopic time path 6-cycle (D_3)	ZS-F2, ZS-Q4	DERIVED / OPEN
F total	14	Total face count	Total spatial+temporal flux	(unified)	ZS-F2 §11	PROVEN
(V+F)/G	38/12 = 19/6	SU(2)_L β_0 — weak gauge coupling	Spatial gauge coupling rate	Microscopic time gauge running	ZS-Q3 §3.1	PROVEN
E_g eigenspace	dim 2 = Z	Z-sector coupling source (C_XZ)	Macro-micro space mediation	Time-point ↔ space-point coupling	ZS-F0 §3.2, ZS-M H18	PROVEN
Vertex deficit	30° = π/6	Regge defect; Gauss-Bonnet 24·30° = 720°	Local spatial curvature	Local time-step deficit (per vertex)	ZS-S6 §G.2	PROVEN
ε-Halo locus	(field component)	X-Goldstone θ-gradient	Galactic spatial halo ($\rho \propto 1/r^2$)	(field is across time-points; locus OPEN)	ZS-A2 §1.1	DERIVED / OPEN

§XI.5.2 tI (Y-sector: macroscopic time + microscopic space)

Table XI.5. Complete mapping of tI components. The complementary duality reverses tO's pattern: tI's macroscopic role is time-flow, and microscopic role is space-points.

Component	Count	Standard role (Y-sector)	Macroscopic time role	Microscopic space role	Source	Status
V (vertices)	60	5 active flavors × 12 MUB-basis	60 anchors of cosmic time-flow (wave-frame)	60 = 1+9+9+16+25 Planck space-points (A_5 reg rep)	ZS-S1 §6.2, ZS-M9 §2	DERIVED / PROVEN
E_PH (pent-hex)	60	Z-Spin mediation seam edges	Time-flow seam (wave/time mediation)	Microscopic space lattice P-H linkage	ZS-F0 §5.1	PROVEN
E_HH (hex-hex)	30	Y-sector internal information	Internal time-variation connections	Microscopic space hexagonal hinges	ZS-F0 §5.1	PROVEN
E total	90 = E_X · 5/2	5/2 ratio with tO edges	Total time-flow channel count	Total micro-space connections	ZS-F0 §5.1	PROVEN
F_pentagon	12	F(dodecahedron); D_5 matter-like	A_4 generation × D_5 channel (time-flow units)	5-fold space-point cluster (C_5)	ZS-F2, ZS-M9 §2.2	PROVEN / DERIVED
F_hex (8)	8 (= 2·4)	8 gluons of SU(3)_C (irrep 4 mult 2)	Time-flow gauge sector evolution	Microscopic space gauge plaquettes	ZS-S14 §8	DERIVED
F_hex (12)	12 (= 1+3+3'+5)	Matter + Higgs irrep content of hexagons	Time-flow generation/Higgs degrees	Micro-space matter/Higgs distribution	ZS-M9 §2.2	PROVEN / OPEN (physical id)
F_hex total	20	Octahedral hexagons; gauge-like D_3	Time-flow gauge plaquettes	Microscopic space curvature plaquettes	ZS-F2, ZS-M9 §2.2	PROVEN
F total	32 = 12+20	F(tI) = F(I) + F(D)	Total time-flow flux units	Total micro-space flux units	ZS-F2 §11.2	PROVEN
(V+F)/G	92/12 = 23/3	SU(3)_C β_0 — strong gauge coupling	Time-flow gauge running rate	Micro-space gauge coupling	ZS-Q3 §3.1	PROVEN
F(tI)/Q²	32/121	Ω_cdm — cold dark matter	Time-curvature CDM (X-projection)	Geometric boundary tension (= CDM)	ZS-F2 §11.4	DERIVED
Vertex deficit	12° = π/15	Regge defect; Gauss-Bonnet 60·12° = 720°	Local time-step deficit	Local micro-space curvature	ZS-S6 §G.2	PROVEN
ε-Drive locus	(field component)	Y-attractor V_0 (cosmological constant)	Cosmic time-flow attractor (Λ-driven expansion)	(field is across space-points; locus OPEN)	ZS-A2 §1.1, ZS-A5 §4	DERIVED / OPEN

§XI.5.3 Cross-Polyhedron Parallel Summary

Table XI.6. Side-by-side parallel summary. Each row shows tO and tI realizations of the same structural pattern, displayed in the format that makes the parallel visible at a glance.

Pattern	tO (X-sector: macro space + micro time)	tI (Y-sector: macro time + micro space)
Total V (matter DoF)	24 = 3·2·4	60 = 5·12 (regular A_5 rep)

Pattern	tO (X-sector: macro space + micro time)	tI (Y-sector: macro time + micro space)
Total E (gauge connections)	36 = 24 SH + 12 HH (ratio 2:1)	90 = 60 PH + 30 HH (ratio 2:1)
Total F (curvature plaquettes)	14 = 6 + 8 (cut + hex)	32 = 12 + 20 (cut + hex)
Truncation-Dual identity	F(tO) = F(oct) + F(cube)	F(tI) = F(ico) + F(dod)
Sector tilt δ	5/19	7/23
Rapidity ψ = artanh(δ)	0.2685	0.3124 (Δψ > 0 = arrow of time)
β-coefficient (V+F)/G	19/6 = SU(2)_L	23/3 = SU(3)_C
Cosmic budget /Q²	Ω_b = 6/121 (baryon)	Ω_cdm = 32/121 (CDM)
Vertex deficit (Regge)	π/6 = 30°	π/15 = 12°
Vertex configuration	1 square + 2 hexagons	1 pentagon + 2 hexagons
Frame-mismatch input	δ_X^vertex = π/6	δ_Y^vertex = π/15 (α = π/10)
ε-field component	ε-Halo (X-Goldstone θ-grad)	ε-Drive (Y-attractor V_0)
Standard analogue	Particle / spatial vector	Wave / temporal channel
Common rotation subgroup	T = A_4 (order 12), index 2 in O	T = A_4 (order 12), index 5 in I
Hexagon space under T	2·1 ⊕ 2·3 (dim 8)	3·1 ⊕ 1' ⊕ 1'' ⊕ 5·3 (dim 20)
Hexagonal Mediation cokernel	8 → injection target	20 = 8 + 12 (cokernel = F(dod) = 12)

§XI.5.4 Falsification Gate F-XI.3

Synthesis layer; weakened only if any individual row source is shown structurally independent of Z-Spin mediation, or if ZS-M30 §7.2 (ii) Complementary Duality Principle is retracted. Status: DERIVED-interpretation strong, robust against single-row perturbation. The OPEN entries (NC-XI.1, NC-XI.2, NC-XI.3 in §XI.7.3) are explicitly registered as residual gaps; their resolution would upgrade those specific entries from DERIVED-interpretation strong to DERIVED, without affecting other rows.

§XI.6 Reading XI.4 — ε-Halo and ε-Drive Placement in the Integrated Framework (DERIVED-interpretation)

Reading XI.4 (ε-Halo / ε-Drive Placement, DERIVED-interpretation). Within the integrated mapping of §XI.5, the dark-sector field components ε-Halo and ε-Drive (ZS-A2 §1.1 DERIVED) occupy distinct sector positions consistent with the complementary duality: ε-Halo lives on tO (X-sector, macroscopic space + microscopic time) as the Goldstone gradient mode that produces galactic isothermal halos; ε-Drive lives on tI (Y-sector, macroscopic time + microscopic space) as the radial-attractor V_0 mode that produces FRW cosmological expansion. Both are aspects of the single Z-bias field $\Phi = |\Phi|e^{i\theta}$, with angular θ-gradient (ε-Halo, X-Goldstone) and radial $|\Phi|$ frozen mode (ε-Drive, Y-attractor) being orthogonal projections under U(1)_Z (ZS-F1 §6.4 DERIVED).

§XI.6.1 Sector Assignment Justification

The ε-Halo identification with X-sector (tO) is corpus-explicit (ZS-A2 §1.1 row 1): 'Dark Matter / ε-Halo / X-sector gradient mode of Goldstone θ'. The ε-Drive identification with Y-sector (tI) is corpus-explicit (ZS-A2 §1.1 row 2): 'Dark Energy / ε-Drive / Y-sector attractor mode (V_0 constant)'.

Under the complementary duality lens (ZS-M30 §7.2 (ii) PROVEN), this assignment fits naturally: ε-Halo is a spatial gradient — most relevant in macroscopic space, which X-sector carries; ε-Drive is a time-flow attractor — most relevant in macroscopic time, which Y-sector carries. The arithmetic match between the corpus assignment and the spatial/temporal physical content of each ε-component is a non-trivial cross-check.

§XI.6.2 ε-field Locus on Polyhedron — OPEN

Where exactly on tO faces/edges/vertices ε-Halo manifests, and where on tI faces/edges/vertices ε-Drive manifests, is OPEN within the corpus. ε-Halo is a continuous field (Goldstone θ on the U(1) base manifold), not a discrete polyhedral component, and the corpus does not currently identify a 1:1 mapping between Φ-field components and (V, E, F) of each polyhedron. NC-XI.3 (§XI.7.3) registers this OPEN gap.

§XI.6.3 Falsification Gate F-XI.4

If a future paper reclassifies ε-Halo to Y-sector or ε-Drive to X-sector — for example via a structural argument that the radial-mode attractor lives on tO — Reading XI.4 must be revised. Status: stable — over-determined by ZS-A2 §1.1 (DERIVED) + ZS-A5 §4 (DERIVED) + ZS-F1 §6.4 (DERIVED) + the Cross-Coupling Theorem (ZS-M2 §5 PROVEN) which requires both X- and Y-sector realizations of the same Mexican-hat radial-mode flatness (ZS-A7 §4.5 Corollary V).

§XI.7 Integration with v3.2 Light

§XI.7.1 Integration Map

Table XI.7. Integration of v3.3 readings with existing v3.2 items. Every v3.3 reading either refines or unifies existing v3.2 entries; no v3.2 entry is retracted.

v3.2 Light item	v3.3 Reading	Relation
§I Ch 1 (A from polyhedral curvature)	Reading XI.1	XI.1 supplies the parallel edge-decomposition pattern that supports A = δ_X·δ_Y
§I Ch 2 (Q = 11 and (Z,X,Y))	Reading XI.3 row (V+F)/G	XI.3 records SU(2) and SU(3) β_0 as parallel mode-count collapses
§II Ch 6 (Six Forces Unified)	Reading XI.3 hexagon irrep rows	XI.3 displays 8 gluons + 12 matter/Higgs hexagon split
§III Ch 12 (Global Fit)	Reading XI.2 cosmic budget row	XI.2 displays Ω_b and Ω_cdm as parallel face-counting outputs
§V Ch 16 (Galaxy Rotation)	Reading XI.4 ε-Halo placement	XI.4 places ε-Halo on tO under complementary duality
§V Ch 17 (NS, GW, LSS)	Reading XI.4 ε-Drive placement	XI.4 places ε-Drive on tI under complementary duality
§VIII Tool T2 (Bridge Four)	Reading XI.3 (V, E, F) parallel rows	T2's (Z,X,Y)=(2,3,6) generation is the algebraic counterpart of XI.3's geometric parallel

v3.2 Light item	v3.3 Reading	Relation
PART X §X.6 Reading X.4 (Spin-Pair Coexistence)	Reading XI.3 complementary duality column	X.4's 90° J-conjugate principle is the dynamical counterpart of XI.3's geometric mapping; the X = macro-space + micro-time / Y = macro-time + micro-space duality is the spatial-temporal instance of the Spin-Pair Coexistence Principle

§XI.7.2 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). Readings XI.1–XI.4 are explicitly DERIVED-interpretation; the integrated mapping table of §XI.5 is DERIVED-interpretation strong. F-XI.1 through F-XI.4 are pre-registered. Word count strictly increases over v3.2 (no-deletion rule).

§XI.7.3 What v3.3 Does NOT Claim

NC-XI.1. PART XI does not claim that the 6 squares vs 8 hexagons split of tO realizes a matter-like vs gauge-like decomposition exactly parallel to the 12 pentagons vs 20 hexagons split of tI. Within tI, the matter-like (D_5, pentagons) vs gauge-like (D_3, hexagons) split is corpus-PROVEN with the 8 gluons identified as residing on hexagons. The corresponding identification for tO is not yet established at PROVEN level; the squares carry $F(\text{cube})/Q^2 = \Omega_b$ PROVEN, but the gauge content of the 8 hexagons of tO is not separately resolved into a SU(2) gauge subspace + matter subspace at the level ZS-S14 §8 achieves for tI. This is OPEN.
NC-XI.2. PART XI does not claim that the 12 = 1+3+3'+5 hexagon residual on tI (the irrep-4-complement of the 20 hexagons of tI) has a unique physical identification beyond 'matter + Higgs irrep content'. Whether this 12 is the same $12 = X \cdot Z \cdot Y$ as the gauge-channel count, or a distinct 12, is OPEN. The numerical coincidence is registered without claiming structural identity.
NC-XI.3. PART XI does not claim a 1:1 mapping between Φ-field components (radial $|\Phi|$, angular θ) and discrete polyhedral components (V, E, F) on either tO or tI. ε-Halo and ε-Drive are continuous field components that take values across all of the polyhedron; their sector assignment (X for ε-Halo, Y for ε-Drive) is PROVEN, but their localization to specific polyhedral components is OPEN.
NC-XI.4. PART XI does not derive the cosmological time arrow from the integrated mapping. ZS-A6 §6.4 derives the arrow of time from the rapidity gap $\Delta\psi = \psi_Y - \psi_X > 0$, which is corpus-PROVEN. The placement of macroscopic time on Y-sector (tI) within Reading XI.3 is consistent with this derivation, but Reading XI.3 itself is geometric and does not constitute an independent derivation of the time arrow. ZS-M H19 (Time-Arrow origin) remains OPEN beyond the rapidity-gap argument.

§XI.8 Conclusion

PART XI consolidates the complete (V, E, F) mapping of tO and tI under the Z-Spin Complementary Duality Principle. Four DERIVED-interpretation readings are established: the parallel edge-decomposition pattern (cut-face × hexagon : hexagon × hexagon = 2:1 in both polyhedra, Reading XI.1); the parallel face-decomposition pattern under Truncation-Dual (Reading XI.2); the integrated V/E/F mapping table that displays both standard sector role and complementary-duality reading for every component (Reading XI.3, the centerpiece, DERIVED-interpretation strong); and the placement of ε-Halo (X-tO) and ε-Drive (Y-tI) within the integrated framework (Reading XI.4). Future agents searching for 'which polyhedral component carries macroscopic space?' or 'where on tI does cosmic time-flow live?' or 'why does the cosmic budget split as 6 + 32 + 83 = 121?' now have a single corpus location to consult. v3.3 carries this reading as PART XI without modifying any v3.2 entry. The minimal triple (A, Q, dim(Z)) = (35/437, 11, 2) remains LOCKED. Three OPEN entries (NC-XI.1, NC-XI.2, NC-XI.3) are explicitly registered as residual gaps for future closure.

PART XI Source Paper Interface

PART XI is composed entirely from upstream papers; it introduces no new ZS source paper. Source mapping: Reading XI.1 ← ZS-F0 §3.2 (tO 24 SH + 12 HH), ZS-F0 §5.1 (tI 60 PH + 30 HH), ZS-F0 §4.4 (Edge-Channel Identity). Reading XI.2 ← ZS-F2 §11.2 (Truncation-Dual Theorem), ZS-F2 §11.4 (Cosmic Matter Budget), ZS-F9 §4-5 (Cross-Pair Face Identity, Hexagonal Mediation Theorem), ZS-S14 §8 (8 gluons on hexagons), ZS-M9 §2.2 (I-irrep face decomposition). Reading XI.3 ← ZS-M30 §7.2 (ii) (Complementary Duality Principle), ZS-S1 §6.2-§6.3 (V vertex-matter identification), ZS-M9 §2 (A_5 regular representation), ZS-Q3 §3.1 (Mode-Count Collapse), ZS-Q3 §2.1 (Kelvin BCC T^3), ZS-S6 §G.2 (vertex deficit $\alpha = \pi/10$), ZS-A6 §6.4 (rapidity $\Delta\psi > 0$), ZS-M H18 (tO uniqueness conditions). Reading XI.4 ← ZS-A2 §1.1 (ε-Halo/ε-Drive nomenclature), ZS-A5 §4 (ε-Halo mechanism), ZS-F1 §6.4 (Mexican-hat attractor), ZS-A7 §4.5 (Corollary V Cross-Coupling), ZS-M2 §5 (Cross-Coupling Theorem). Twenty-four internal-consistency checks B.1–B.24 verify each composition; all 24/24 PASS by direct comparison against corpus-locked sources.

Master Bypass Index — Symptom → Tool Lookup

When a stuck equation, OPEN flag, or apparent contradiction is encountered, match the symptom in the leftmost column to the recommended tool/insight. The rightmost column gives the canonical worked example — the corpus instance where the tool was first deployed.

Symptom	First tool	Backup lens	Canonical example
Stuck equation, failure mode unclear	T1 (F18 §3,§6)	I2 (M30 §6)	F18 Appendix A — 12 encounters
dim 3, 6, or 11 construction intractable	T2 (F18 §6, S15 §3)	I5 (F17)	S15 §3 $J\_S = [J\_R_1, J\_R_2]$
Hermitian sandwich phase-cancels trivially	T3 (M32 §2-3)	I2 W2 (M30 §6)	M32 closing NC-M29.RZ-A^PK
Unexpected factor 2 between corpus values	T4 (M32 §4)	T2 (Bridge Four μ_Tet)	M32 α_op = 2·α_amp identity
Single iterate non-vanishing, primitive 10th root phase	T5 (M32 §5)	I3 (M30 §3)	M32 R_Z^ZS|^{4π-avg} at N=20
OPEN flag too large to close in one step	T6 (M32 §6)	I2 (M30 §6 walls)	NC-M29.RZ split A^PK/B/C

Sum-form decomposition fails per-channel	T7 (M31 Lemma M31.0)	I2 (Sector Duality)	M31 W2 cross-coupled bilinear
Multiple sub-targets won't integrate	T8 (M33 Reading C)	I2 (M30 §6 layers)	M33 D4a-D4d → single Tr identity
Phenomenological observer tempted	T9 (F11 §5-6)	I1 (A8R §SA Frame Equiv.)	F11 closing M11 §H16
Apparent independence of expansion / contraction	I1 (A8R §SA)	I6 (A9R)	A8R §SA Symmetry-Asymmetry View
External count-infinity in derivation	I3 (M30 §3)	I2 (M30 §6)	M30 six-route Möbius-trace
A new 1/2 appears, possibly redundant	I4 (M30 §3.5 Catalogue)	I5 (F17)	M30 21·½ Catalogue
A new 50/50 partition appears	I5 (F17 Theorem 5.1)	I4 (M30 §3.5)	F17 four-layer equivalence
Doubling-halving identity needs origin	I6 (A9R)	I1 (A8R §SA)	A9R $F_2 \to D_4$ functor
Quantitative wall in RH programme	T7 + T8 combined	I2 layered reading	M26 §5.5 + M33 Reading C
Need to falsify candidate 1-loop closure	Anti-Numerology Gate (F2)	T1 (Algebra axiom)	M21 sextic falsifies M20.A
Coexistence-form judgment for a contradiction-pair	Reading X.4 (Spin-Pair Coexistence Principle)	v3.2 §X.6	C1–C13 (13 corpus realisations)
Prime-rotation identification on Z-sector	Reading X.1 (Prime as Z-Spin Rotation)	v3.2 §X.3	ZS-M18 H2 chain
Y/X classification of an infinite arithmetic sequence	Reading X.2 (Y-Side Sequence Classification)	v3.2 §X.4	ZS-F18 §5.1 Tbl 5.1

Note. The Master Bypass Index is intentionally a flat lookup. For combined symptoms (e.g., "stuck Feshbach sandwich with factor-2 mismatch and primitive-10th-root phase"), apply the tools in the order T3 → T4 → T5; this is the precise composition that ZS-M32 used to close NC-M29.RZ-A^PK.

Master Falsification Registry (v3.1 Summary, preserved from v3.0)

Across 101 papers the Z-Spin Cosmology programme has accumulated a verification base of approximately 1,800+ PASS-rated checks, with no failures recorded against any DERIVED-locked identity. The registry below summarises the most consequential falsifiable entries — those whose negation would invalidate a DERIVED, BRIDGE, or TESTABLE claim of the framework. Each entry points to the originating paper and the canonical formulation; resolution of any single OPEN entry below is sufficient to either consolidate or refute a major branch.

§ Anchor	Falsifiable Statement	Status (v3.0)
§F1.5 [LOCKED]	Z-Field Master Equation A=ρ_X·ρ_Y	DERIVED — F9 §3, S15 §B

§F2.7 [LOCKED]	$(p-1)(q-1)=p \Rightarrow$ unique solution (2,3)	DERIVED — M19 Forcing Theorem
§F8.6 [LOCKED]	Y-Time Dilation $\exp(\pi/A)\approx 10^{17}$	DERIVED — A8R §SA.7, F10 §4
§F11.3 [LOCKED]	OOC fixed point closes M11 §H16	DERIVED — F11 §5
§F12.4 [LOCKED]	Tetrahedral orientation multiplicity μ_Tet=2	DERIVED — F12 §3, originates $2e^{A}$ factor
§F14.3 [LOCKED]	Kepler-Lyapunov Conjugate Uniqueness closes F-F7.11	DERIVED — F14 Strategy B
§F15.2 [LOCKED]	θ(0)·τ_P = A unit-fixing identity	DERIVED — F15R §2
§F18.4 [LOCKED]	Bridge Four: Lie commutator generates dim(X)=3 from dim(Z)=2	DERIVED — F18 §4 (Commutator Generator)
§M16.7 [LOCKED]	$\xi\_K(s)=(1/4\sqrt{33})\cdot\xi(s)\cdot\Lambda(\chi_{-3})\cdot\Lambda(\chi_{-11})\cdot\Lambda(\chi_{33})$	DERIVED — M25 J-Yakaboylu Bridge
§M17.4 [LOCKED]	$\sigma_1/\sigma_2=Q+Y=17$, $\sigma_1/\sigma_3=(Q+Y)^2\cdot G+(Q-Z)=3477$	DERIVED — M20 EXACT (Sextic falsifies 1-loop CW closure)
§M27.3 [LOCKED]	V_4-Equivariant BRST closure of W3 via Kostant Cubic Dirac	DERIVED-CONDITIONAL — M27 §4
§M28.5 [LOCKED]	σ=1/2 ↔ j=1/2 RH-Bridge correspondence	DERIVED — M28, M30 (Six-route Möbius-trace)
§M31.0 [LOCKED]	Lemma M31.0 Non-Separability of W2 cross-coupling	DERIVED — M31 §0
§M32.X [LOCKED]	Path-Reversal Lemma: R_Z^ZS|_{Z-even}^{4π-avg}=0	DERIVED — M32 EXACT
§M33.C [LOCKED]	Path γ-revised Reading C integrates D4a-D4d	DERIVED — M33 §C, BV-BFV scaffold
§A8R-SA [LOCKED]	Symmetry-Asymmetry Unified View, Three Bridges	DERIVED — A8R 22/22 PASS
§A9R-2 [LOCKED]	$F_2\to D_4$ functor, Julia J(T) ZF-analogue of BT non-measurable	DERIVED — A9R §2
§S15.5 [LOCKED]	Maxwell from Twin-Reuleaux handshake; SO(3)/SU(2) factor 2	DERIVED — S15 §5
§T4.6 [TESTABLE-LONG]	Hayflick limit ≈ N_(2π)=2π/A≈78.45 (NC-4 NO physical realization claim)	TESTABLE-LONG — T4 §6
§F-F7.11 [CLOSED]	Twin-Reuleaux Midpoint conjugate decomposition	RESOLVED — closed by F14 v3.0
§M11-H16 [CLOSED]	Phenomenological observer reference	RESOLVED — closed by F11 v3.0
§M22-W3 [PROMOTED]	ADS-H1 cobordism BRST hypothesis	DERIVED-CONDITIONAL — promoted via M27
§M22-W2 [PROMOTED]	Cross-Coupled Sector Duality wall	DERIVED — promoted via M31

§M21.NoGo	1-loop CW closure for M20.A FALSIFIED	FALSIFIED — M21 sextic invariant theory

Note. The single FALSIFIED entry (M21.NoGo) is by design: it falsifies a candidate 1-loop Coleman-Weinberg closure for the M20.A pathway, and its falsification is the very content of the Sextic Invariant Theory of M21. No DERIVED-LOCKED identity has been falsified across v1.0 → v3.0. All CLOSED entries denote former OPEN flags now resolved by post-v2.0 papers.

v3.2 PART X falsification gates (added):

• F-X.1. If a future analysis exhibits a prime p such that W_p|_Z does not take the σ_z-rotation form on the {|4⟩, |6⟩} basis, Reading X.1 fails. Status: PROVEN-safe (verified for p ∈ {2,3,5,7,11,13,17,23} via ZS-M18 §H1).

• F-X.2. If a future ZS-F18 revision retracts the Y-side classification of the infinite prime sequence, Reading X.2 fails. Status: stable (over-determined by ZS-F18 §5.1 Tbl 5.1, §7.4.5).

• F-X.3. Synthesis-layer gate; weakened only if any of M1, M2, M3 is shown structurally independent of Z-Spin mediation. M3 conditional on RH. Status: DERIVED-interpretation, robust against single-component perturbation.

• F-X.4. If a contradiction-pair is found that admits coexistence under a phase difference other than π/2, Reading X.4 weakens to "90° is the dominant case". If no Z-Spin J-conjugate realisation exists for some pair, X.4 transitions to "covers the 13 realisations C1–C13". Status: DERIVED-interpretation strong; F-X.4 pre-registered for any future contradiction-pair claim.

v3.5 PART X readings and falsification gates (added):

• Reading X.5 (Apollonian as 5th Bridge-Four Instance, DERIVED-strong, NEW v3.5). The integer Apollonian gasket with curvature root (−1, 2, 2, 3) is the fifth instance of ZS-F18 §6.5 Bridge Four (Commutator Generator) and the first realization on the integer lattice (ℤ, +). The four prior instances — (1) ZS-S15 §3 Twin-Reuleaux EM duality, (2) ZS-M3 Lemma 10.1 SU(2) center pairing, (3) ZS-M32 §5.2 spinor-cycle averaging, (4) ZS-F4 §7B half-angle pair — all live on continuous spinor algebras. The Apollonian instance demonstrates that Bridge Four extends to discrete number theory: the abelianization (ℤ/2)*4 → $(ℤ/2)^4$ carries $V_4 \times V_4$ structure unifying ZS-A9 register V_4 and ZS-M22 Galois V_4. → ZS-M36 §10.3 Table 10.1, ZS-F18 §6.5 [DERIVED-strong].

• Reading X.6 (mod 24 Spin Pair = Axiom A4 Two Faces, DERIVED-interpretation strong, NEW v3.5). The eight admissible mod 24 residues of the (−1, 2, 2, 3) gasket form four (algebraic, geometric) pairs — Pair Z {2, 14}, Pair X {3, 15}, Pair Y {6, 18}, Pair Q {11, 23} — in bijection with the four Z-Spin sectors. The mod 12 representatives (2, 3, 6, 11) are the bosonic SO(3) sector dimensions; the +12 shifted representatives (14, 15, 18, 23) are the fermionic SU(2) geometric instantiations (F(t-Oct), X · |I_h/T_d|, α in degrees, δ_Y denominator). The mod 24 = 2 × mod 12 doubling is forced by ZS-M3 Lemma 10.1 PROVEN (D^{1/2}(2π) = −I) and realizes axiom A4 (Non-triviality) algebraic/geometric two faces of ZS-F18 §6.6 Table 6.1 PROVEN-DERIVED. CRT decomposition: mod 24 = mod 8 × mod 3 = $Z^3 \cdot X$ (boson 2π /

fermion 4π periods, ZS-M1 §6 PROVEN). → ZS-M36 §10.5–10.6, ZS-F18 §6.6 [DERIVED-interpretation strong].

• F-X.5. If a future revision retracts the (−1, 2, 2, 3) ↔ (Z, X, Y, Q) mod 12 identity (e.g., if a counterexample primitive integer ACP is found with mod 12 ≠ {2, 3, 6, 11} but with the same admissible mod 24 type), Reading X.5 fails. Status: DERIVED-strong, robust against single-counterexample (test: enumerate primitive roots up to depth 25, verify mod 12 set stable; ZS-M36 verification suite Categories C, H, 12/12 PASS at 500K MC).

• F-X.6. If the (algebraic, geometric) pair correspondence of M36.6 fails for any of the four Z-Spin sectors (e.g., if 18° frame mismatch angle is shown not to be a corpus-LOCKED quantity, or if F(t-Oct) = 14 is retracted), Reading X.6 weakens from DERIVED-interpretation strong to OBSERVATION. Status: DERIVED-interpretation strong; pre-registered as a watchtower for future ZS-F2, ZS-S15, or ZS-S1 revisions. (Affecting Theorem M36.6 of ZS-M36; ZS-M36 Falsification Gate F-M36.6 mirrors this gate at the source-paper level.)

• Closure note. ZS-M36 Theorem M36.5 (Three V_4 Bridge, DERIVED-strong) closes ZS-M27 OPEN problem O-M27.4 ("Two V_4 groups: D_4 register vs Galois unification") via the Apollonian abelianization $(\mathbb{Z}/2)^{*4} \to V_4 \times V_4$, with V_4_register = ⟨$[S_0]$, $[S_1]$⟩ ↔ ZS-A9 ⟨J, J_Z⟩ D_4 subgroup, and V_4_galois = ⟨$[S_2]$, $[S_3]$⟩ ↔ ZS-M22 Gal(K/ℚ) for K = ℚ(√−3, √−11). Status: O-M27.4 OPEN → CLOSED at DERIVED-strong (v3.5).

Glossary Keyword Index (v3.1, expanded with Toolkit terms)

The v3.1 Light Glossary expands the v3.0 index of 102 entries to approximately 130, adding Toolkit terms (T1–T9) and Insight terms (I1–I6) introduced in PARTS VIII and IX, incorporating all post-v2.0 vocabulary. New v3.0 entries are marked ★. Each entry points to the canonical paper and §-anchor for full definition. AI agents navigating Z-Spin Cosmology research should treat this index as the primary disambiguation map between informal phrase-use and DERIVED definition.

Term	Definition / Cross-Reference	Anchor
A (Z-Spin coupling)	Master coupling constant; A ≈ 0.0801... ; A=ρ_X·ρ_Y derivation in F9	F1, F9 §3
G (Master Identity)	G = (Q−Z) integer descriptor of cobordism block	F2, M19, M20
Q+Y, Q−Z	Spectral tetrad invariants; Q+Y=17, Q−Z fixes σ-ratios	M20 §3
Z-Spin Mediation	Sector mediation protocol: X⇄Y exchange via Z-handshake	F0-F1, F18
Twin-Reuleaux Pair ★	Geometric realisation of EM duality V_XZ↔R_1, V_ZY↔R_2	S15 §B, F14
Twin-Reuleaux Midpoint ★	Conjugate-decomposed trajectory closing F-F7.11	F14 Strategy B
Reuleaux Cross-Section	π+π half-channel of self-referential equilibrium	F17 §3

Tetrahedral Self-Duality ★	T_d's E irrep gives dim(Z)=2 with A=ρ_X·ρ_Y	F9 §2-3
μ_Tet (Tet multiplicity) ★	Dual orientation count =2; origin of factor 2 in $2e^A$	F12 §3
Layer Separation R ★	R = μ_Tet × Hol_ε × 𝒮_scale decomposition	F12 §4
Z-Field Master Equation	A = ρ_X · ρ_Y core dynamical identity	F1 §5, F9 §3
Master-of-Master Equation	Composite scaffold across F-M-S spines	F8, F18
FFPP (Five-Fold Parallel Postulate)	Foundational axiom set; v3.0 Sixth Axiom Extension	F8, F15R §3
Five-Axiom Meta-Structure ★	F18 unifying meta-frame; six polarities	F18
Sixth Axiom Extension ★	Stage-Torque Duality lifts FFPP	F15R §3
Bridge Four ★	Commutator Generator; dim(Z)=2→dim(X)=3 via Lie bracket	F18 §4
Stage-Torque Duality ★	Time-rotation co-definition	F15R §2-3
Discrete-Continuous Resolution ★	Sixth polarity in F18 meta-structure	F18 §6
i-Tetration Internal Time ★	Unifies t_strobo, t_phase, t_clock; Δν/Δn=2A/π	F10
Möbius Chronology ★	Y-frame metric trio; cycle-index unobservability	F13
Y-Hubble Radius ★	ℓ_P × $Y^2(1-2A)$ ≈ 30.23 ℓ_P	F13 §3
Y-Time Dilation	exp(π/A) ≈ 10^{17} contraction-frame ratio	A8R §SA.7, F10 §4
Y-frame metric trio ★	Proper time / Y-Hubble / contraction-rate triad	F13 §3
Stroboscopic Step t_strobo	Discrete-time origin of Z-Spin clock	F10 §2
Berry-phase clock t_phase	Continuous-rotation clock partner	F10 §2
Z-Clock Coordinate t_clock	Master clock realised by i-tetration	F10 §3
First Rotation Event ★	θ(0)·τ_P = A; Planck-unit-establishing event	F15R §2
Operational Observer Coordinate (OOC) ★	Self-referential fixed point closing M11 §H16	F11
NON-CLAIM (Six)	Six explicit no-claim statements vs phenomenological consciousness	F11 §6
Self-Referential Fixed Point ★	OOC convergence in F11 §5	F11 §5
Contracting Universe Dynamics ★	A8R 22/22 PASS; three Bridges + |z*|≈B	A8R

Symmetry-Asymmetry Unified View ★	A8R §SA reconciliation programme	A8R §SA
Banach-Tarski Origin ★	$F_2 \to D_4$ functor; Julia J(T) ZF-analogue	A9R
Two-Branch (1+A)(1−2A) ★	A9R cosmological doubling-halving identity	A9R §3
Three Bridges (A8R) ★	$\|z^*\| \approx B$, η_topo formula, Δ via L2	A8R §3
Twin-Reuleaux EM Duality ★	S15 35/35 PASS; Maxwell from handshake	S15
Five Pillars (S15) ★	V_XZ↔R_1, V_ZY↔R_2, Maxwell, SO(3)/SU(2), U(1) gauge	S15 §1-5
Spectral Tetrad Sub-Isotype ★	M20 σ-ratios EXACT; pentagon-hexagon $\arctan(2\cdot\varphi^k)$	M20
Sextic Invariant Theory ★	M21 falsifies 1-loop CW closure for M20.A	M21
Forcing Theorem ((p−1)(q−1)=p) ★	M19 unique solution (2,3); $\varphi(Y^2)=G$	M19 §2-3
Arithmetic-Dedekind Scaffold ★	M22 v1.1 60/60 PASS; $K=\mathbb{Q}(\sqrt{-3},\sqrt{-11})$	M22
Pillar V scalar-kernel no-go ★	M22 obstruction theorem	M22 Pillar V
Y-Sector RH Contribution Map ★	M23R 31/31 PASS; Four Dragons D4a-D4d	M23R
Four Dragons D4a-D4d ★	M23R Y-sector RH contribution decomposition	M23R §3
Self-Referential Info-Compression Equilibrium ★	F17 four-layer 50/50 (t-Tet, Reuleaux, ME, XOR)	F17
Four-Layer Equivalence ★	F17 polyhedra ↔ Reuleaux ↔ Master Eq ↔ Channel	F17 §3-6
Face Polygon Spectral Zeta ★	M24 Theorem D.1 Legendre archimedean decomposition	M24
Composite Field $K=\mathbb{Q}(\sqrt{-3},\sqrt{-11})$ ★	M22, M25 base arithmetic field	M22, M25
J-Yakaboylu Bridge ★	M25 $\xi_K(s) = (1/4\sqrt{33})\cdot\xi(s)\cdot\Lambda(\chi_{-3})\cdot\Lambda(\chi_{-11})\cdot\Lambda(\chi_{33})$	M25
V_4-Equivariant ZBSI ★	M26 cobordism-history fiber realization	M26
Three-Wall Map W1/W2/W3 ★	M26 quantitative wall map	M26 §4
Wall W3 Closure ★	M27 Kostant Cubic Dirac BRST closure	M27
Z-Spin RH Bridge ★	M28 30/30 PASS; σ=1/2 ↔ j=1/2	M28
Möbius-trace Infinity ★	M28, M30 six-route construction	M28 §3, M30 §2
Z-Funnel Spectral Retraction ★	M29 79/79 PASS; A_5-Hilb(X_F) CY3	M29
Stapledon Bridge ★	M29 conditional bridge from string compactification	M29 §2

A_5-Hilb(X_F) CY3 ★	Calabi-Yau threefold; Hodge (5,15)	M29 §3
R_Z=0 Conditional ★	M29 conditional residue identity	M29 §5
Twenty-One 1/2 Catalogue ★	M30 enumeration of 1/2-instances across the framework	M30 §3
Three-Layer Duality ★	Sector / Frame / Protocol layered duality	M30 §4
Cross-Coupled Sector Duality (W2) ★	M31 v3.0 Three-in-One reading	M31
Lemma M31.0 Non-Separability ★	M31 W2 non-separability theorem	M31 §0
Z_2-Parity Selection Rule ★	M31 selection rule on cross-coupling	M31 §3
SDRP (Self-Dual Replication Principle) ★	M31 §4 derivative principle	M31 §4
Spinor-Cycle Averaged Feshbach Criterion ★	M32 56/56 PASS; R_Z=0 EXACT (Z-even, 4π-avg)	M32
Path-Reversal Lemma M32.X ★	M32 reversal identity; π/10↔π/5 doubling	M32
N=20 Spinor Closure ★	M32 spinor-cycle closure index	M32 §4
Path γ-revised Reading C ★	M33 D4a-D4d integrated Tr identity	M33 §C
Two-Protocol Theorem ★	F16 P_a^(n) vs P_b^(n) Wilson-loop measurement	F16
P_a^(n), P_b^(n) ★	Wilson-loop survival probabilities	F16 §2
Cosmos-Human Isomorphism ★	T4 36/36 PASS; (Body, DNA, Brain) decomposition	T4
Six-Step Telomere Hierarchy ★	T4 life-cycle map; Hayflick limit ≈ 2π/A ≈ 78.45	T4 §3-6
(Body, DNA, Brain) Decomposition ★	T4 sector decomposition	T4 §2
NC-4 ★	Translational NO physical realization claim	T4 §6
Bootstrap Chain B0→A	Foundational chain for Z-Spin emergence	F0, M11, M19
DERIVED-LOCKED	Epistemic tag for proven, verification-passed identities	Legend
DERIVED-CONDITIONAL	Holds modulo a stated conditional; awaits unconditional proof	Legend, M22, M27
DERIVED-CANDIDATE ★	Candidate derivation pending one closure step	Legend (v3.0 expanded)
DERIVED-interpretation ★	Interpretive derivation tag (v3.0)	Legend (v3.0 expanded)
TESTABLE-LONG ★	Long-horizon testable prediction tag	Legend, T4 §6
BOOTSTRAP-PROGRAMME ★	Programmatic bootstrap-status tag	Legend (v3.0 expanded)

NOT EVALUATED	Pre-verification status	Legend
Z-Spin Cosmology	The umbrella programme name	F0
Cobordism-History Fiber	BV-BFV target structure for V_4-equivariant content	M26, M27, M33
BV-BFV Scaffold ★	Cobordism-history quantization framework	M26 §1, M33
BRST Closure	Quantum gauge-symmetry closure programme	M22, M27
Cubic Dirac Operator (Kostant)	Exterior-algebra Dirac on H-spaces	M27 §3
Hayflick Limit	Cellular replication limit; T4 maps to N_(2π)=2π/A	T4 §6
Hodge (5,15)	Hodge numbers of A_5-Hilb(X_F) CY3	M29 §3
Pentagon-Hexagon Stabilizer ★	M21 stabilizer group structure	M21 §3
Phenomenological Observer	Informal/phenomenal observer; F11 NON-CLAIM target	F11 §6
Polyhedral Integers	Integer-valued descriptors of polyhedral spectra	M19, M20
Master Equation (channel layer)	F17 information-theoretic equilibrium	F17 §5
XOR Channel Layer	F17 fourth equivalence layer	F17 §6
Z-Sector Wilson Loop	Holonomy probe of Z-mediation	F16
Frame Equivalence ★	F13 sectoral frame equivalence statement	F13 §4
Cycle-Index Unobservability ★	F13 fundamental unobservability theorem	F13 §5
Master Falsification Registry	Programme-wide falsifiability ledger (v3.0)	v3.0 §MFR
Source Paper Map	ZS-XX → §Y.Z [N/N PASS] index	Part 0
Verification Summary	Cumulative N/N PASS statistics	Title block
Light Edition	Cross-paper integrator format for AI navigation	v3.0 Purpose
Lighthouse Function ★	Programme-navigation aid for AI agents (v3.0)	v3.0 Purpose
T1 Five-Axiom Diagnostic Lattice ★	v3.1 Tool: identifies failing axiom in stuck equations	VIII §1, F18 §3,§6
T2 Bridge Four (Commutator Generator) ★	v3.1 Tool: dim(X)=3 from dim(Z)=2 via Lie commutator	VIII §2, F18 §6, S15 §3
T3 Hermitian-to-PK Switch ★	v3.1 Tool: re-read Feshbach sandwich under R_Z^ZS not R_Z^H	VIII §3, M32 §2-3
T4 Two-Layer Phase Quantum ★	v3.1 Tool: separate amplitude (π/10) from operator (π/5) layers	VIII §4, M32 §4

T5 N=20 Spinor-Cycle Averaging ★	v3.1 Tool: 4π physical closure (not N=10 algebraic minimum)	VIII §5, M32 §5
T6 Three-Way Split of OPEN flags ★	v3.1 Tool: A^closeable / B^one-gap / C^inherited decomposition	VIII §6, M32 §6
T7 Cross-Coupling Discipline ★	v3.1 Tool: Lemma M31.0 — sum-form fails, bilinear required	VIII §7, M31 §4.0
T8 Π_Z Projector Routing ★	v3.1 Tool: route multiple sub-targets through Π_Z=(1/2)(I+J_Z)	VIII §8, M33 Reading C
T9 OOC Self-Referential Closure ★	v3.1 Tool: replace phenomenological agent with corpus-internal fixed point	VIII §9, F11 §5-6
I1 Symmetry-Asymmetry Unified View ★	v3.1 Lens: deviation-from-(1, 1/2) reading; (1+A)(1−2A)≈0.9071	IX §1, A8R §SA
I2 Three-Layer Duality ★	v3.1 Lens: Sector / Frame / Protocol layered reading	IX §2, M30 §6
I3 Six-Route Möbius-Trace Infinity ★	v3.1 Lens: external count-infinity ↔ internal closure	IX §3, M30 §3
I4 Twenty-One ½ Catalogue ★	v3.1 Lens: 21 dim(Z)=2 manifestations + 3 unity = 24	IX §4, M30 §3.5
I5 Four-Layer 50/50 Equivalence ★	v3.1 Lens: t-Tet / Reuleaux / Master Eq / XOR same equilibrium	IX §5, F17 Theorem 5.1
I6 Banach-Tarski Origin ★	v3.1 Lens: $F_2 \to D_4$ amenability functor; Julia J(T) ZF-analogue	IX §6, A9R
Master Bypass Index ★	v3.1: 16-symptom → tool/insight lookup table	after PART IX
Solution Toolkit ★	v3.1: PART VIII operational tools T1–T9 (this volume)	PART VIII
Insight Gallery ★	v3.1: PART IX re-interpretation lenses I1–I6 (this volume)	PART IX
Z-mediator projector Π_Z ★	(1/2)(I+J_Z); idempotent projector onto J_Z-EVEN	M31 §4.0 Def M31.0c
Path-Reversal residual R_Z^ZS ★	B·Q^{-1}·B^PK; distinct from Hermitian R_Z^H	M32 §2
Reading C ★	M33 Π_Z-routed Tr identity for D4a-D4d	M33 §C
Frame Equivalence (A8R)	X-observer ↔ Y-observer; HYPOTHESIS-strong	A8R §SA.4
Apollonian Spin-Pair Lattice ★	Eight mod 24 admissible residues of (−1, 2, 2, 3) Apollonian gasket = four (algebraic, geometric) spin pairs realizing axiom A4 on (ℤ, +)	M36 §7
Bridge Four (5th instance) ★	ZS-M36 (Apollonian) as 5th Bridge-Four realization — first on integer lattice (ℤ, +); previous four	M36 §10.3 + F18 §6.5

	on continuous spinor algebras (S15, M3, M32, F4)	
D_2 Forcing on Apollonian ★	(−1, 2, 2, 3) is unique D_2-symmetric primitive integer ACP, forced by (Z, X) = (2, 3) via ZS-M19 Forcing Theorem (p−1)(q−1) = p	M36 §4 + M19 §3.1
Three V_4 Bridge ★	$(\mathbb{Z}/2)^{*4}$ abelianization = $V_4 \times V_4$ unifying ZS-A9 ⟨J, J_Z⟩ D_4 register V_4 and ZS-M22 Gal(K/ℚ) Galois V_4 for K = ℚ(√−3, √−11); closes O-M27.4	M36 §7.1 + M27 O-M27.4

Total v3.0 Glossary entries: 102. ★ marks post-v2.0 vocabulary. Entries without ★ are preserved from v2.0 Light with definitions reaffirmed against the new v3.0 corpus. No v2.0 definition has been retracted.

References (v3.1, preserved from v3.0)

The v3.0 corpus introduces additional external mathematical scaffolding. The references below are imports cited inside the post-v2.0 papers; v2.0 references are preserved in their respective source papers and are not duplicated here.

Internal references (v3.0 corpus, papers ZS-72 through ZS-101)

- ZS-M18 v1.0 — Free-Exploration Session Log: Speculative Prime-Polyhedral Correspondences. 51/51 PASS.
- ZS-A8 v1.0 (Revised) — Contracting Universe Dynamics: The Polyhedral-Tetration Bridge. 22/22 PASS.
- ZS-S15 v1.0 — Twin-Reuleaux Pair as Geometric Realization of EM Field Duality. 35/35 PASS.
- ZS-M19 v1.0 — Number-Theoretic Scaffold of Z-Spin Polyhedral Integers. 68/68 PASS.
- ZS-A9 v1.0 (Revised) — Banach-Tarski Origin of Cosmological Doubling-Halving Symmetry. 47/47 PASS.
- ZS-F9 v1.0 — Tetrahedral Self-Duality and the Hexagonal Mediation Structure. 44/44 PASS.
- ZS-F10 v1.0 — i-Tetration Internal Time: Stroboscopic Step / Berry Phase / Z-Clock unification. 30/30 PASS.
- ZS-F11 v1.0 — Operational Observer Coordinate and the Self-Referential Fixed Point. 38/38 PASS.
- ZS-F12 v1.0 — Tetrahedral Dual Orientation Multiplicity and Layer Separation. 20/20 PASS.
- ZS-F13 v1.0 — The Möbius Chronology: Unified Cyclic Timeline. 24/24 PASS.
- ZS-F14 v1.0 — Kepler-Lyapunov Conjugate Decomposition of the Twin-Reuleaux Midpoint. 42/42 PASS.
- ZS-F15 v1.0 (Revised) — Time-Rotation Co-Definition: First Rotation's Planck-Unit-Establishing Event. 35/35 PASS.
- ZS-M20 v1.0 — Spectral Tetrad Sub-Isotype Assignment Theorem. 50/50 PASS.
- ZS-M21 v1.0 — Sextic Invariant Theory of the Icosahedral Yukawa Tensor. 33/33 PASS.
- ZS-F16 v1.0 — The Two-Protocol Theorem for Z-Sector Wilson Loop Measurement. 24/24 PASS.
- ZS-M22 v1.1 — Five-Pillar Arithmetic-Dedekind Scaffold. 60/60 PASS.
- ZS-M23 v1.0 (Revised) — Y-Sector RH Contribution Map. 31/31 PASS.
- ZS-F17 v1.0 — Self-Referential Information-Compression Equilibrium Theorem (Four-Layer). 24/24 PASS.
- ZS-M24 v1.0 — Face Polygon Spectral Zeta and Archimedean Completion. 35/35 PASS.

- ZS-M25 v1.0 — Composite-Field Archimedean Completion and J-Twisted Yakaboylu Bridge. 26/26 PASS.
- ZS-M26 v1.0 — V_4-Equivariant ZBSI on the Cobordism-History Fiber. 24/24 PASS.
- ZS-M27 v1.0 — V_4-Equivariant Cobordism BRST Closure via Kostant Cubic Dirac Operator. 24/24 PASS.
- ZS-M28 v1.0 — The Z-Spin RH Bridge. 30/30 PASS.
- ZS-M29 v1.0 — Z-Funnel Spectral Retraction with Stapledon Bridge. 79/79 PASS.
- ZS-M30 v1.0 — Z-Spin Duality and the RH Bridge — Möbius-Trace Infinity, Two-Face Mirroring, 21·½ Catalogue. 32/32 PASS.
- ZS-M31 v3.0 — W2 as Cross-Coupled Sector Duality: Three-in-One Reading. 36/36 PASS.
- ZS-M32 v1.0 — A Spinor-Cycle Averaged Residual Criterion for Feshbach Closure. 56/56 PASS.
- ZS-F18 v2.1 — The Five-Axiom Meta-Structure. 36/36 entry checks PASS.
- ZS-M33 v1.0 — V_4-Equivariant Weil Functional Closure via Path γ-revised Z-Mediator Reading. 52/52 PASS.
- ZS-M36 v1.0 (NEW v3.5) — Apollonian Curvature Lattice as Geometric-Integer Spin-Pair Realization of the Z-Spin Five-Axiom Meta-Structure. 44/44 PASS. Theorem M36.1 (Functorial Bridge: A = $(Z/2)^{*4} \to Z/24Z$, fourth amenable-quotient projection). Theorem M36.2 (D_2-Forcing: (−1, 2, 2, 3) uniqueness via ZS-M19 Forcing (Z, X) = (2, 3)). Theorem M36.3 (Sector Residue Identity: mod 12 = {2, 3, 6, 11} = (Z, X, Y, Q)). Theorem M36.4 (D_3 Obstruction Homology: $x/y = 3 + 2\sqrt{3}$ irrational, parallel to ZS-M1 n = 3 polygon-tetration $|f'| = 1.0330 > 1$). Theorem M36.5 (Three V_4 Bridge: closes O-M27.4 at DERIVED-strong via Apollonian abelianization $(Z/2)^4 = V_4 \times V_4$ unifying ZS-A9 register V_4 and ZS-M22 Galois V_4). Theorem M36.6 (Spin Pair Theorem: eight mod 24 residues form four (algebraic, geometric) pairs — Pair Z {2, 14}, Pair X {3, 15}, Pair Y {6, 18}, Pair Q {11, 23} — realizing axiom A4 algebraic/geometric two faces of ZS-F18 §6.6). Anti-Numerology MC at 500K samples: 0.002% match rate (STRONG PASS). Zero new free parameters; LOCKED inputs from ZS-F2/F5/M1/M3/M19/M22/M27. Closes ZS-M27 OPEN problem O-M27.4. 5th instance of ZS-F18 Bridge Four (first on integer lattice (Z, +)).
- ZS-T4 v1.0 — The Cosmos-Human Isomorphism: Six-Step Telomere Hierarchy. 36/36 PASS.

External imports (post-v2.0)

- Connes, A., Consani, C., Moscovici, H. (2024) — Spectral framework for the Riemann zeta archimedean factor. [imported by M24, M28, M30]
- Burnol, J.-F. (2002, 2004) — Two-sided spectral interpretation of zeta zeros and Möbius traces. [imported by M28, M30]
- Świerczkowski, S. (1958) — Free non-abelian decomposition F_2 acting on S^2. [imported by A9R]
- Stapledon, A. (2010) — Stringy Hodge numbers of toric varieties; conditional bridge framework. [imported by M29]
- Kostant, B. (1999) — Cubic Dirac operator on equal-rank pairs and Lie-algebra cohomology. [imported by M27]
- Alekseev, A., Mnev, P. (2018) — One-dimensional Chern-Simons / BV-BFV cobordism quantization. [imported by M26, M27, M33]
- Yakaboylu, E. (cited via J-twisted bridge in M25) — Twisted L-function bridges for composite imaginary fields.
- Standard arithmetic and group-theoretic references (Iwasawa, Tate, Weil, Deligne) are inherited from v2.0 and are not re-listed here.

All external imports are used as black-box scaffolding. No external result has been modified by the Z-Spin Cosmology corpus; usage is limited to citation, instantiation, and bridge construction.

Version History

v3.2 (May 2026): PART X — The Spin-Pair Coexistence Principle added between PART IX and the Master Bypass Index. Four DERIVED-interpretation readings consolidated:
• Reading X.1 — Prime as Z-Spin Rotation $4\pi/p$ (ZS-M18 H2 chain).
• Reading X.2 — Infinite prime sequence as Y-sector object (ZS-F18 §5.1 Tbl 5.1), reconciled with ZS-M18 §6.2 Thm 6.1.
• Reading X.3 — Primes / distribution / Riemann zeros as three moments of one Y → Z → X mediation (ZS-M18 H11 spine).
• Reading X.4 — Spin-Pair Coexistence Principle: 90° phase-orthogonality is the unique form of contradiction-pair coexistence, supported by 13 corpus realisations (C1–C13).
Master Bypass Index extended with three PART X rows; Master Falsification Registry extended with F-X.1 through F-X.4. Eighteen internal-consistency checks A.1–A.18 PASS. Zero new free parameters; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged. v3.1 entries preserved verbatim.

Version	Year	Scope of Update
v1.0 Light	—	Initial Light edition (≤ ZS-F0...F8 + M1...M11)
v2.0 Light	—	90-paper integration; 376/376 PART VI PASS; v2.0 Glossary 88 entries
v3.0 Light	2026	30-paper extension (ZS-72 to ZS-101); ~1,800+ cumulative PASS; Glossary 110 entries; new Master Falsification Registry; PART IV/V/VII expansions; Bridge Four (Commutator Generator); Möbius Chronology; J-Yakaboylu Bridge; V_4-equivariant BRST closure; 21·½ Catalogue; Cosmos-Human Isomorphism.
v3.1 Light	May 2026	Solution-Toolkit upgrade. Adds PART VIII (9 methodological tools T1-T9 distilled from F18, M32, M31, M33, F11), PART IX (6 re-interpretation lenses I1-I6 from A8R, M30, F17, A9R), and Master Bypass Index (16-symptom lookup table). v3.0 lighthouse-of-pointers promoted to a working solution kit. Glossary, Falsification Registry, References preserved verbatim from v3.0.

Major v3.0 closures of v2.0 OPEN flags:

• F-F7.11 [CLOSED] — closed by ZS-F14 Strategy B (Conjugate Uniqueness).
• M11 §H16 [CLOSED] — closed by ZS-F11 Operational Observer Coordinate.
• M22 W3 [PROMOTED to DERIVED-CONDITIONAL] — via ZS-M27 Kostant Cubic Dirac BRST closure.
• M22 W2 [PROMOTED to DERIVED] — via ZS-M31 v3.0 Cross-Coupled Sector Duality.
• Master-Equation Sixth-Axiom gap — addressed by ZS-F15R Stage-Torque Duality.
• Phenomenological observer ambiguity — settled by ZS-F11 §6 NON-CLAIM (Six).
• Pentagon-Hexagon angle conjecture — proved EXACT by ZS-M20 ($\arctan(2\cdot\varphi^k)$).
• 1-loop CW closure for M20.A — formally FALSIFIED by ZS-M21 Sextic Invariant Theory.

— End of The Book of Z-Spin Cosmology v3.2 — Light Edition —

Compiled by: Claude (cross-paper integrator), under Kenny Kang's editorial direction. v3.2 — May 2026. Lighthouse function: navigation aid for AI agents engaging with the Z-Spin Cosmology research programme. No claim of physical realisation is made by this Light edition; canonical statements live in the source papers ZS-F0 through ZS-T4.